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Driving Weight Management

A Framework for Optimized Weight Prediction in Automotive Engineering

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ABSTRACT

Today, the automotive industry faces complexity in managing vehicle configuration processes as the need for product customization increases. This thesis investigates the current weight management system used at Volvo Cars, with a focus on improving the abstraction processes of vehicle weight during configuration. Current processes are dependent on manual effort, expert judgment and complex variant expressions in the product data system. These limitations result in inefficiencies and inaccuracies in the weight management processes.

Through a case study approach, including qualitative interviews with internal stakeholders, key pain points within current practices were identified. Combined with a literature search on product configuration, weight and variant management, and configuration systems, a framework approach has been presented. The framework aims at increasing the accuracy and efficiency of the weight management system, while reducing the need for manual labor. This is done through a proposed algorithm logic for automating the assignment of part weight to variant layer partial configurations which utilize existing product structures.

The result contributes to the automotive industry with a better understanding of the challenges in weight management when dealing with high customization. Further, the results suggest actionable improvements that can support Volvo Cars in achieving automation in vehicle configuration processes.

KEYWORDS
Product customization, vehicle configuration, variant management, weight management, weight prediction.

SAMMANFATTNING

Fordonsindustrin står idag inför en ökande komplexitet i hanteringen av fordonskonfiguration i takt med att efterfrågan på produktanpassning växer. Detta examensarbete undersöker det nuvarande systemet som används för viktprediktering på Volvo Cars, med särskilt fokus på att förbättra abstraheringen av viktdata under konfigurationsprocessen. Nuvarande arbetssätt är starkt beroende av manuellt arbete, expertbedömningar och komplexa variantstrukturer i produktdatasystemet. Dessa begränsningar leder till ineffektivitet och bristande precision i viktberäkningarna.

Genom en fallstudie med kvalitativa intervjuer med interna experter har centrala problemområden i dagens processer identifierats. I kombination med en litteraturstudie inom produktkonfiguration, vikt- och varianthantering samt konfigurationssystem, presenteras ett ramverksförslag. Ramverket syftar till att öka precisionen och effektiviteten i viktpredikteringen, samt att minska behovet av manuellt arbete. Detta görs genom en föreslagen algoritmlogik som automatiserar tilldelningen av artikelvikt till partiella konfigurationer inom variantlagret. Existerande produktstrukturer utnyttjas för att uppnå detta.

Resultatet bidrar till en ökad förståelse för de utmaningar som viktprediktering innebär i samband med ett ökat behov för hög produktvariation. Vidare föreslås konkreta förbättringar som kan stödja Volvo Cars i att implementera automation i konfigurationsprocessen.

NYCKELORD
Produktanpassning, fordonskonfiguration, varianthantering, vikthantering, viktprediktering.

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LIST OF DEFINITIONS

BOM – Bill of Materials: A comprehensive list of parts and quantities necessary for the manufacturing of a sellable product.

Complete configuration: A finalized product configuration where compatible and acceptable choices of variants have been made. Also referred to as vehicle variant.

Configuration rules: The set of constraints set on a configuration system to ensure valid complete configurations.

IUR – Item Usage Rule: The steering which outlines for which variant or variant combination a specific part applies and should be used in production. Also referred to as variant expression.

Product configuration: The process of selecting available features resulting in a desirable and sellable product.

Partial configuration: A part of a product configuration where compatible and acceptable choices of variants have been made. Can be expanded to at least one complete configuration.

PLM – Product Lifecycle Management: The organization, system and practice a company uses to handle data and information of a product throughout its life cycle. It is an integrating system, combining several aspects of a firm.

Variant: A distinct version of a part or feature, differing in geometry, material or color.

Variant families: A grouping of variants with similar characteristics which are mutually exclusive in a complete product configuration.

Weight abstraction: The assignment of weight from the item layer to the feature layer in a two-layer configuration system.

Weight delta: A discrepancy between predicted weight, calculated in the weight prediction, and actual weight, measured at the end of the production line.

Weight management: The system of tracking and handling weight data in a PLM and product development process.

Weight prediction: The calculation of the weight of a vehicle prior to production.

1 INTRODUCTION

In this chapter, the main purpose of this paper is presented. Along with a short introduction to the topic, the context and motivations behind the project are introduced. Lastly, a problem statement and the delimitations that apply for this project are described.

1.1 Background

Today's consumers increasingly expect products tailored to their unique preferences and needs, driving companies to embrace product customization. This demand has caused a large increase of mass customization, which is defined by Hvam et al. (2008) as "a production form in which customized products are delivered while exploiting the advantages of mass production." Mass customization has completely changed the ways of working with product design and manufacturing by enabling businesses to offer a wider range of options. By selecting from a range of available parts and features, customers can achieve a customizable product, this process is known as *product configuration*. Companies need to rely on tools such as advanced manufacturing technologies and intelligent product configuration systems to successfully achieve this. These tools make it possible for firms to efficiently manage the complexity of customization while also meeting customer expectations for personalized products at scale. In the automotive industry, product configuration is essential to meet specific customer needs in a global market. Factors such as local regulations, cost, and production time drive these variations, resulting in different end products. Further, customer preferences for powertrains, engines, packages and options also drive variance in the product range. The resulting amount of possible finished products can be immense. Managing this complexity is critical for businesses and is considered a key differentiator in the competitive automotive market (Rock et al., 2015).

To handle product configurations throughout processes and divisions within a firm, *Product Lifecycle Management* (PLM) systems are used, integrating information from all parts of the organization throughout the lifetime of the product (Corallo et al., 2013). As an important property of parts and assemblies, weight is included in the PLM systems of vehicles. Together with other important properties, it is considered a key characteristic and should be individually labeled as such, requiring special attention and evaluation (Zheng et al., 2008). The processes and methods used for tracking and evaluating the weight of a vehicle are known as *weight management*. For automotive manufacturers offering customizable vehicles configured in sales configurators, the weight of the vehicle changes depending on its unique configuration. The practice of calculating a vehicle’s weight based on said configuration is known as *weight prediction* (Kidambi et al., 2014). Variations in customized products, based partly on previously mentioned customer choices on powertrains, engines, packages and options, result in a large variance in total weight as well as weight distribution on front and rear axles. Accurate weight prediction is therefore a critical aspect in the PLM processes, as it directly affects factors such as performance, compliance and production. During the development process, the total weight and weight distribution on front and rear axles need to be calculated based entirely on its configuration as a vehicle has not been produced yet, showing the need for accurate weight prediction.

The occurrence of discrepancies between predicted weight, measured in kg and found in the weight prediction, and actual weight, measured at the end of the production line, are referred to as a *weight delta* and may result in significant costs. The costs for the firm arise from different avenues of the organization, with both the inaccuracy of the prediction and the work done to correct weight deltas that

has occurred. This motivates and gives context to the necessary work of finding root causes to discrepancies and developing processes which aim to decrease the occurrence of weight deltas.

The weight management system at Volvo Cars is an integral part of the PLM system and plays a critical role in calculating vehicle weight and axle load distribution during the configuration process. However, the current system is constrained by its reliance on complex, manual processes and data that is often incomplete or subject to frequent changes during development. These limitations result in inefficiencies and inaccuracies that affect several areas of product development and documentation, as well as the production pipeline. The current system results in an excessive amount of manual labor, slowing down the weight management process and increasing the risk of inaccuracy. With significantly more than millions of possible vehicle configurations, the process becomes time intensive and error-prone, especially when design changes are introduced at a later stage of the product development process.

Accurate weight management is essential for product performance, production and compliance with regulation. The current approach struggles to meet these demands efficiently, leading to increased production costs, reduced accuracy, and suboptimal resource utilization.

1.2 Purpose and Objectives

The purpose of the thesis is to investigate the problems with the current weight management system at Volvo Car Group, referred to as Volvo Cars throughout this report. Initially, this entails identifying pain points and root causes. The objective is to suggest a solution that will improve the accuracy and efficiency of the prediction system, while reducing the manual work required to maintain the system.

1.3 Delimitation

The proposed solution focuses on improvements and adaptations which address root causes of discrepancies in weight delta. It does not include the development of processes which require changes in the current product structures of Volvo Cars, as these are deemed unlikely to be implemented. Moreover, the solution does not include improvements in processes which are a result of a weight delta, as these are seen as non-value adding. Further, the solution addresses the weight system for a given timeframe and does not incorporate the dynamic nature of the product throughout its lifecycle. Any calculations or estimations on cost aspects related to the project are also not addressed due to the limited time frame.

2 FRAME OF REFERENCE

This chapter presents the theoretical foundation and key concepts related to weight management and product configuration within the automotive industry. Best practices and existing research on the topic are also presented. Finally, a summary of findings is presented along with the research questions that this thesis work was based on.

2.1 Overview of Weight Management Systems

The weight management system of a firm handles all aspects of weight throughout the lifecycle of the product. This includes the establishment, documentation and usage of weight data for a product's specific purpose.

2.1.1 Weight Considerations for Automotive Industry

In the automotive industry, a key characteristic of any part or assembly is its weight (Zheng et al., 2008). A key characteristic is defined as "a feature of a material, process, or part/assembly whose variation has a significant influence on product fit, performance, service life, or manufacturability" (SAE International, 2017). Weight should therefore have special considerations throughout the product development process, facilitating accuracy and flow of information. Successful management of key characteristics lowers the variability in manufacturing, resulting in benefits such as decreased defects, increased product performance and lower manufacturing costs (Zheng et al., 2008).

The accuracy of weight information for automotive vehicles is important for customers, as there are regulations on total vehicle weight, as well as specific tax implications based on vehicle weight (Skatteverket, 2024; Transportstyrelsen, n.d.). Further, weight also impacts fuel consumption for vehicles with combustion engines and range for electric vehicles (Carlson et al., 2013).

2.1.2 Product Lifecycle Management

Weight management falls under the umbrella of a firm's product lifecycle management, PLM. Such systems, methods and practices are used to integrate product information in an organization throughout its lifetime, from concept development to end-of-life. The PLM systems of a firm are also responsible for ensuring information access and consistency across the organization. A successful implementation of PLM systems thus enables cross-functional collaboration regarding people and processes (Corallo et al., 2013). For complex products, such as automotive vehicles, with a high number of parts and configurations, this implementation becomes very important.

The parts of the vehicles are managed within the firm's *Product Data Management*, or PDM system, which is a part of the overall PLM. Within this system is the *Bill of Materials*, or BOM, which facilitates successful production. The BOM outlines the parts and quantities that make up a finished product. Each part can be either a single unit or the result of a sub-assembly of components, which also is outlined in the BOM. Further, the BOM contains necessary characteristics and requirements of the included parts (Chung & Fischer, 1994). For automotive manufacturers, the BOM is specific for the individual vehicle, informing production of the specific parts and placements for that vehicle. Here, a specific structure can be used, called a *generic BOM* (Tidstam, 2014). Such a structure handles multiple variations of the same product by incorporating rules and dependencies that define which parts are applicable under different conditions.

2.1.3 Computer Aided Design

As an integrated part of automotive PLM systems, Computer Aided Design, or CAD, is used to design parts and assemblies. The implementation of CAD software has transformed product development by allowing data transfer from and between functions within a firm, significantly decreasing design time (Sharma et al., 2014). Firstly, the collaborative aspect of CAD, which has benefited the product development process of automotive vehicles. Secondly, by providing a geometrical product definition, a CAD-model can be created, which combines parts and assemblies to create a virtual model of a configured vehicle. This further benefits the product development process in the automotive industry, as it allows for simulation and calculations regarding vehicles. These simulations and calculations are for instance related to the aerodynamics and weight of the vehicle (Mario, et al., 2013).

2.2 Product Configuration in Automotive Engineering

Today, product configuration enables companies to offer tailored products based on customer preferences. This is especially common and important in industries who deal with complex products, such as the automotive sector. Here, configuration involves selecting predefined parts, such as engine types and interior options, while still ensuring that the result is technically feasible, manufacturable, and compliant with regulations.

2.2.1 The Role of Product Configuration in Vehicle Development

Involving customers in the design process leads to more personalized and satisfying solutions, strengthening their connection to the product and brand itself, shown in a case by Franke and Piller (2004). For manufacturers, customization creates a competitive advantage as it provides information about customer preferences while reducing inventory risks, such as overstocking or product shortages. Additionally, customers are often willing to pay more for tailored solutions, resulting in increased profitability. The desirable possibility for customers to personalize products requires that manufacturers work with configuration processes to handle the challenges that customization brings. According to Mittal & Frayman (1989), configuration is a special case of design activity where a product or system is assembled from a set of predefined parts that can only be combined in specific ways based on constraints and rules. That means configuration is not an open-ended design process but rather a controlled assembly of predefined elements into a valid product.

The automotive industry relies heavily on product configuration to manage the increasing complexity of modern vehicle production. As highlighted by Tidstam (2014), vehicles are highly configurable products, where each vehicle is assembled from a set of predefined parts that together must fulfill technical and regulatory constraints. And when dealing with mass customization, where one vehicle model can generate millions of possible variations, it is necessary to efficiently manage these rules and constraints. In those cases, configuration tools help to ensure correct function and effective production for each of the possible end-products. The challenge is to ensure that all parts for each possible customization function together, while at the same time maintaining the ability to efficiently manage the vast number of possible configurations.

The level of complexity that customization brings makes manual handling impractical, requiring advanced configuration systems to support the design and production process. To handle this complexity in a systematic way, Hvam et al. (2008) indicate the importance of structured configuration models. Their research explains that successful configuration systems must fulfill the following:

- Compatibility: Every selected part must be compatible with other parts in the configuration, ensuring the final product functions correctly.
- Handle constraints: Technical, legal, and production constraints must be supported by configuration systems. The systems also need to hinder invalid or unwanted combinations that could lead to safety issues or inefficiencies.
- Production efficiency: By effective structuring of configuration, manufacturers can optimize supply chain logistics, inventory management, and assembly processes, reducing both lead time and costs.
- Product structures: a standardized and modular product structure helps manufacturers handle large product variation while maintaining efficiency in both design and manufacturing.

By implementing configurations systems that fulfill the requirements mentioned above, the selection of valid product variations can be highly automated. Such systems help to ensure that only feasible and manufacturable combinations are allowed to be created, resulting in minimized errors and inefficiencies. Further on, Hvam et al. (2008) highlights the wanted outcomes from an effective and successful configuration system. Firstly, the need for manual effort is reduced because of the automated systems, which improve accuracy and reduce engineering workload. Secondly, integration and collaboration between manufacturing, engineering, product design and sales is supported, resulting in a seamless transition from customer specifications to production. Lastly, scalability can be enabled if the systems manage to adapt to future technological advancements and product variations. As a conclusion, implementation of rule-based and structured configuration systems enables manufacturers to design and produce vehicles in a cost effective and scalable manner as they also efficiently can manage the increasing customer customization requests.

2.2.2 Configuration Systems and Structure of the Generic BOM

As the complexity of customization increases, more advanced configuration systems are required. According to Tidstam (2014), such systems could either be open or closed, handling product variety with different purposes. Open configuration systems allow for a freer type of part selection with minimal predefined constraints. This type enables high flexibility when it comes to customization, but it also increases the complexity when ensuring that all selected parts are compatible and manufacturable. An example of open configuration is the generic BOM, which enables manufacturers to achieve flexible yet controlled customization. Open configuration systems are also commonly used to manage highly configurable products, such as specialized industrial machinery or luxury vehicles. Closed configuration systems, on the other hand, restrict the feature choices to a fixed set of predefined options, which ensures better efficiency and consistency in production compared to using the more open configuration systems. This approach is commonly used in mass production, where standardization is necessary **to** maintain quality and to handle assembly processes.

Assuring a correct combination of parts for a specific vehicle is critical. The product configuration process evaluates each customer selected option to determine that the vehicle and its features are correct and feasible. Each of these selections corresponds to a *variant*, representing a distinct version of a part, features or systems. According to Anderson (1997), variants can be divided into either external or internal variety when handling configurable products. External variety is perceived by customers as product differences, while internal variety instead is based on the usage of different parts within the production system. For example, external variety can be shown as optional features while internal variety can be based on regional requirements. The purpose of using variants is to maximize customization options for customers, while minimizing complexity in manufacturing and therefore ensuring balance between flexibility and cost in an efficient way.

Tidstam (2014) highlights how in closed configuration, the products managed are often represented by one single product, for example one BOM or one specific predefined version of a vehicle. As the number of possible product combinations increases, it is no longer possible to define a single product for each possible combination. Therefore, variants are grouped after features into clusters or families, referred to as feature families or variables. These types of variants describe variations in parts and are handled by the generic BOM. The specific variations a feature family or variable can take the form of are called feature variants or values. The kind of model that handles these variants is often referred to as *two-layer configuration systems*. In such configuration systems, the configuration logic is divided into two layers, called the feature domain and the item domain. The feature domain includes the high-level customer choices that are feature based, such as seat heating and power train. The item domain instead represents the actual physical parts which are included or excluded from the final configuration, depending on which feature variants that are selected in the sales configurator. *Item Usage Rules,* or *IURs,* connect these two domains, ensuring that the correct parts are activated based on the selected feature combinations. This separation helps manage complexity and supports the scalable operations that are related to weight management. Variants are managed within the configuration system by beforehand defined constraints, referred to as constraints or *configuration rules*, assuring that only combinations that are technically feasible can be generated. For example, a specific suspension system may only be available with certain powertrains, or if a customer selects a panoramic sunroof, the system must also ensure that the correct reinforcement structures are included for this option to maintain vehicle safety and rigidity.

Voronov (2013) discusses this general structure of variables, values and constraints as a constraint satisfaction problem, or CSP. Further, he defines a valid assignment, referred to in this report as a *complete configuration*, as an assignment of a value from each variable, such that all constraints are respected. A partial instantiation, referred to in this report as a *partial configuration*, is an assignment of values to variables which can be extended to at least one valid assignment, or complete configuration. Voronov further discusses existing general algorithms for solving CSPs, which can find if partial configurations can be extended to complete configuration and thus see if they are valid. A further aspect of algorithms which solve CSPs are their possibility to find all possible solutions for a given set of variables, referred to in this report as its *configuration space*.

A single vehicle model can result in millions of possible configurations, meaning that an open and dynamic configuration approach is vital in the automotive industry today. Tidstam (2014) further explains that instead of manually executing separate BOMs for every possible complete configuration, the system automatically generates the correct parts list based on the selected features, while at the same time ensuring compliance with manufacturing, technical and regulatory requirements. By utilizing the product structure in this way, manufacturers can efficiently handle large scale customization while maintaining scalability, production efficiency, and product quality.

2.2.3 The Link between Product Configuration and Weight Management

As earlier mentioned, by varying the shape, geometry, design or material of parts in vehicles, manufacturers create different parts to fulfill aspects such as customer needs, regulatory requirements and cost constraints. In these processes, it is a challenge to ensure that the weight for each part is assigned correctly in the generic BOM. According to Weber (2009), misplacement or incorrect weight data in the systems could lead to inaccuracies in several aspects of vehicle development, such as weight management. Optimization and suspension adjustments are also affected by weight management data. Crucial aspects such as the center of gravity and axle load distribution of the vehicles are all determined

by weight data. Incorrect information in the generic BOM can therefore lead to misaligned or incorrect suspension design for the unique configured vehicles. Moreover, manufacturing processes are also affected by errors in weight management. Planning of material selection, part reinforcement and production lines are dependent on accurate knowledge and predictions of vehicle weight. If the weight information is not accurately entered in the generic BOM, manufacturers risk incorrect dimensioning of parts, which would lead to increased costs, material waste, or in worst case even compromised vehicle performance.

2.2.4 Challenges in Managing Highly Configurable Vehicles

The activities of managing highly configurable vehicles come with significant technical and organizational challenges for manufacturers within the automotive industry. Ensuring efficient production, product compatibility and correct weight management becomes more complicated as the number of possible vehicle variants increases. Key challenges from activities within configuration of vehicles are related to management of variants, ensuring compatibility, as well as managing areas such as data, product structure, constraints and late-stage design changes. As presented in the previous section, managing the large amount of data that the extensive number of possible configurations generates is challenging. The challenge mainly lays within ensuring that each configuration remains valid, manufacturable and correctly managed in the generic BOM. Ensuring compatibility between the parts is also dependent on well-structured configuration rules to avoid invalid combinations (Tidstam, 2014). Without effective compatibility validation, errors can occur that lead to unwanted consequences such as production delays, increased engineering time and rework costs, and increased risk of post-production recalls (Cannas et al., 2020).

In their study on product configurators, Cannas et al. (2020) discusses the balance between managing generic BOMs that are detailed enough to correctly track the dependencies of each part, while at the same time avoiding a high level of data that is difficult to manage. If the data is of a too detailed level, the system processing will be slowed down, which further on leads to inefficiencies in manufacturing (Otten et al., 2019). But on the other hand, if the level of the generic BOM is too abstract, there is a risk of losing traceability of the specific configurations, according to Tidstam (2014). This would in turn lead to difficulties when managing parts and specific features for each unique car, meaning that the calculations of weight distribution also become more complex. Furthermore, design modifications carried out at a late stage of the product development process may not be reflected in a generic BOM of too abstract level, leading to inconsistencies.

As the number of variants and dependencies increase in modern vehicle configuration systems, so does the computational load. The complexity that comes with these systems requires real-time data processing, calculations and validation to ensure that each configuration chosen by the customer is manufacturable and technically feasible. The main challenges of this aspect, except valid generic BOM structures and configuration selections, are related to the integration with other business systems (Cannas et al., 2020). Interactions between these systems require synchronization of data, and if not optimized, the real-time responses could be slowed down.

2.3 Existing Research and Best Practices

There are several structural approaches available for managing highly configurable vehicles, such as feature trees, modularization, platforms and part families. Commonly used frameworks provide structured ways to handle the complexity that comes with variant selection, modular structures and other types of functional groupings.

2.3.1 Strategies for Variant Management

The purpose of using variations of parts is that it allows manufacturers to alter the design of vehicles and create unique combinations based on customer preferences, without having to entirely create new models each time. Effective variant management in a two-layer configuration system is essential for achieving customization when dealing with highly configurable products. According to Blecker & Abdelkafi (2006), successful variant management strategies can reduce redundant complexity while still ensuring that customer needs are met efficiently. Based on the strategies presented in their paper, successful grouping of variants can be achieved in different ways, which could be categorized as below.

- Part families: where grouping is based on parts similarity, either in terms of design features or manufacturing processes. Redundant parts are minimized, which simplifies inventory management and fosters configuration flexibility.
- Commonality: when parts are being reused across variants, as frequently as possible. This approach reduces costs in both design and production aspects, and it supports assembly, lead-times, inventory management and maintenance.
- Product modularity: builds on standardized interfaces in design, that enables parts to be changed or modified independently, without having to do a full redesign. Such structure ensures greater compatibility between parts, that allows for more efficient customization and reduced production complexity with minimal engineering effort.
- Platform-based: is a strategy that structures parts into platforms with common architecture, and it functions as a common module that is recurring in multiple variants. Shared platforms enable scalability, efficient production and greater flexibility towards meeting market demands.

Moreover, Blecker & Abdelkafi (2006) explain that, based on the structure of the firm and the products managed, the firm should find a suitable variant management strategy. The choice depends on factors such as manufacturing scale, complexity of products, customization preferences and cost efficiency. Aligning variant strategies with the goal of the business can result in achieving stronger competitiveness, agility and savings in cost.

Reiser and Weber (2007) present another method to structure, visualize and manage product variations by using multi-level feature trees for handling complex product families. This model groups and separates variants into platform (high-level) product features or into part (low-level) technical details. This approach helps to ensure dependencies and constraints between the different levels of the vehicle configurations. Such feature-trees also help with the traceability between features and physical parts in the generic BOM of complex vehicles. A clearer separation of feature families ensures better management of relationships between physical parts and functional requirements. Similarly, Tidstam (2014) presents how feature trees can be used as a model for organizing selectable features of a configurable product. The model is hierarchical and built up as a tree-like diagram, as shown in Figure 1. Each node in the tree represents an attribute, earlier referred to as variable or feature family, and its different branches represent the available choices of feature variants or values.

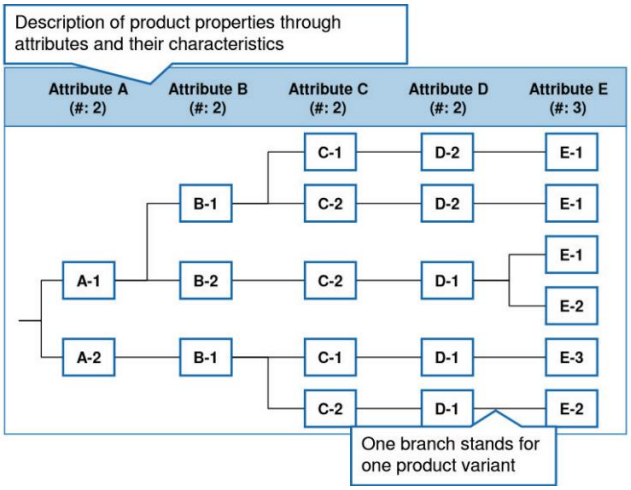


Figure 1. Feature-tree diagram, visualizing the configuration options within features families (Tidstam, 2014).

The feature tree visually captures the dependencies and options that come with configurations, making it easier to understand relationships between features and configuration rules.

2.3.2 Strategies for Product Configuration

The processes of product configuration can be more efficient if supported by structured decision-making methods. The Defense Acquisition University (2001) presents an established methodology called IDEF0 (Integration Definition for Function Modeling), common in manufacturing and process modelling. The method is used to analyze, document, and visualize complex engineering processes in a hierarchical way. Primarily, the model is built up from the two key elements functions (boxes) and data or objects (arrows) that define their relationships. Illustrated in Figure 2, each arrow and its relation to the function box indicates the specific role within the process. Operational inputs are related to operations such as data or objects that are being processed, and the output is supported and regulated by mechanisms (bottom) and control inputs (top). The clarity in processes and decision-making that this structure provides makes it an efficient tool for modeling complex engineering systems.

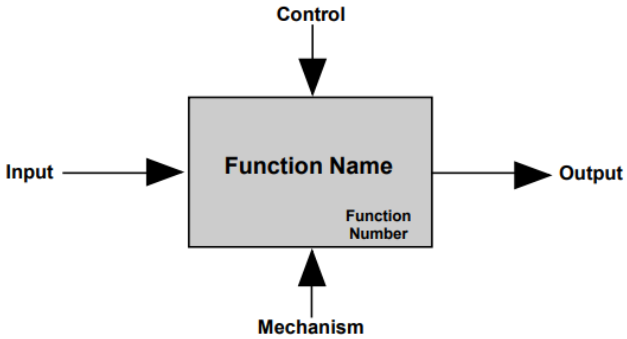


Figure 2. The basic structure of the modelling method IDEF0 (Defense Acquisition University, 2001).

Further on, there are different approaches available for managing product configuration within a two-level product structure. These information systems can be divided into three areas: case-based, model-based, and rule-based configuration. Rule-based configuration uses logical rules to determine which technical features used in manufacturing and engineering apply based on the selected customer features from a sales configurator (Voronov, 2013; Tidstam, 2014). These systems, as described by Hayes-Roth (1985), use rules on the form (*if* condition *then* consequence) to define what actions should be taken to

create a valid configuration. The structure of the *if-then* conditions can for example look like the following:

> *If* all-wheel drive (AWD) is selected, *then* use transmission type A

Typically, the rules connect customer feature variants with the corresponding technical feature variants within the feature domain. The same logic of *if-then* is used in IURs when selecting valid parts from the item domain based on the included feature variants in a complete configuration, where the condition contains technical feature variants, and the consequence contains a part. Conditions can also be evaluated in a fixed sequence, which breaks down the configuration process into a series of steps. Then the rule represents one technical feature family, and each step represents one case where a feature variant is applicable, as well as help determine in what order the actions should be carried out. Each step includes rules structured in a sequential *else-if* logic such as:

> *if* condition *then* consequence
> *else if* condition *then* consequence
> *else* consequence

This method supports how certain actions should take place in relation to others, helping with logic and management. It also enables the sales configuration process to remain intuitive for the customer, while the underlying system incrementally expands the configuration into detailed variants and parts required for engineering and manufacturing purposes.

2.4 Research Questions

The literature has given context and insights to which systems and processes are related to the management of weight within the automotive industry. The increasing demand for customization in the market has resulted in more complex product configurations, leading to challenges for manufacturers when aiming to manage product data. Considerations such as requirements, computational load and data detail must all be balanced to meet external needs as well as internal efficiency. The weight management process becomes more challenging due to this complexity, while also adding to it by requiring additional definition in the systems.

Based on the frame of reference above, more knowledge was needed to identify opportunities for weight management improvements. Thus, three research questions were brought forward to understand problem areas and possibilities for improvement within the process. The questions are related to consequences, challenges and opportunities for weight management within the automotive industry and are stated as follows:

- *What are the consequences of faulty weight prediction in an automotive manufacturer?*
- *What challenges inhibit an efficient weight prediction process in an automotive manufacturer?*
- *How can processes regarding weight prediction for unique vehicle configurations be refined to ensure greater efficiency and accuracy?*

3 METHOD

This section presents the processes carried out during the thesis project. The data collection approach and how data was analyzed is also described, along with a description of the framework development process. Lastly, the validation of the framework and a method criticism is presented.

The thesis followed an applied research approach, and the work was grounded in real-world conditions, focusing on a specific organizational context and practice-oriented problem assigned by Volvo Cars. The study adopts an exploratory and qualitative design, allowing for open investigation into challenges, structures and workflows of the current weight management process at the firm. Positioned between theory and practice, the study aimed to combine academic knowledge with organizational expertise to generate actionable results. This section gives a general overview of the structure of the completed study, visualized in Figure 3.

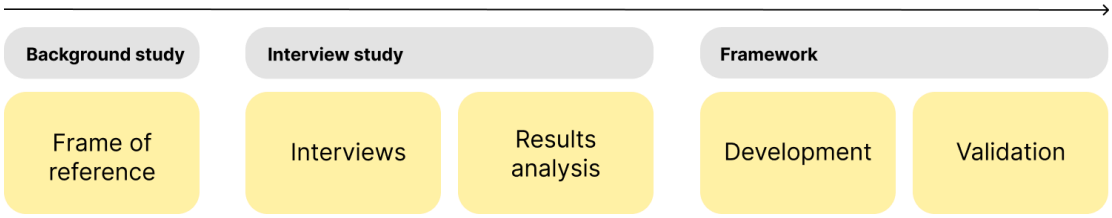


Figure 3. General overview of the thesis methodology.

3.1 Background Study

The purpose of the background study was to outline the frame of reference for the project, setting the appropriate context and necessary background for the empirical study to take place. Based on the frame of reference, together with the aspects outlined in the purpose and objectives found in section 1.2, the problem definition and research questions were developed.

The literature readings provided a foundation for understanding key concepts relevant to the thesis work, such as product configuration, PLM systems and existing strategies for variant management. Databases such as Scopus, Web of Science and Google Scholar were used as the foundation to find relevant literature, where the main keywords of the study (product customization, vehicle configuration, variant management, weight management and weight prediction) were used as a base of the search. Academic papers found were then studied and used to establish a frame of reference, which supported both the problem analysis and the development of the proposed framework. The review also helped identify best practices in managing complexity in configurable products and guided the foundation of interview themes.

3.2 Interview Study

The interview study was completed qualitatively, with semi-structured interviews conducted with internal stakeholders to the studied processes. The subjects were chosen based on relevance and availability together with the industry supervisor. While literature gave a theoretical base, it was not sufficient to get an understanding of the firm-specific and complex nature of Volvo Cars' weight management process. To fill this gap, a set of semi-structured interviews were conducted with process

expert employees within the organization. In total, seven interviews were held with representatives from different departments, all involved in or affected by the weight system in some type of way.

As the weight management process extends over several departments, data had to be collected from most relevant areas to get a clear picture of the current processes and their weaknesses. Interviewees were selected from departments that are either directly involved in weight management or are closely linked to the systems and processes that affect it. Firstly, interviews were held with departments responsible for product documentation and system interfaces. These helped to grasp the overall process and logic behind working with vehicle configuration. To understand the context of the weight system, roles central to maintaining and improving the current weight system were questioned. Furthermore, the part release department was selected to provide insight into how data and changes are managed and implemented in PLM systems. Together, these departments represent key functions across the product documentation chain, all of which helped with the understanding of the cross functional challenges and data flows within the current weight management process.

The interview structure was chosen based on the need to explore a complex and partially undocumented process, while still maintaining the flexibility to adapt to each interviewee's own insights and area of expertise. Based on Adams' (2015) handbook, the semi-structured approach encouraged the interviewees to speak their mind and ideas, opening for discussions in a less stiff meeting environment. Given the rather unstructured nature of the current weight management process, this format was particularly useful in uncovering tacit knowledge, operational challenges, and the dependencies within the different departments. It allowed for adaptation of the discussions, helped to reflect on the unique knowledge and responsibilities of each respondent and helped to map out the core processes.

An interview guide was developed and used based on themes identified during the literature study. It focused on general questions related to the weight management process, product structure, current problem areas and a discussion of possible solutions or wishes related to identified problem areas. This guide was then adapted for each interview depending on the subject's role, department, and relevance to specific aspects of the weight management process. The interview guide in its full can be found in Appendix A. From each interview, a transcript and a recording was taken to allow for verifying of details when analyzing the gathered data. The interviews ranged from 45-80 minutes and were either held in the company office or through a video meeting. A list of each interview and details on subjects, number of respondents, their titles and departments, the focus area of the interview and its time duration can be found in Table 1.

Table 1. Summary of the interviews held for the data collection of internal processes at Volvo Cars.

Subject (number of interviewees)	Role	Department	Focus area	Location	Time (min)	Date
Subject 1 (one)	Solutions Architect	PLM (Product Structure)	Overall PLM process	Teams meeting	60	2025-02-18
Subject 2 (two)	Product Specification Specialist	Product Documentation	Product structure	Company office	50	2025-02-25
Subject 3 (one)	Change Order (CO) Specialist	Product Documentation	Vehicle design	Company office	80	2025-02-25
Subject 4 (two)	Principal Engineer Suspension	Hardware Architecture Department	Vehicle design	Company office	60	2025-02-26
Subject 5 (one)	Software Architect	PLM (Product Structure)	Overall PLM process	Company office	45	2025-02-26
Subject 6 (one)	Tech Lead	Weight Management (Operations)	Weight management process	Company office	60	2025-02-27
Subject 7 (two)	Engineering Manager	Weight Management (Digital)	Overall weight management processes	Teams meeting	60	2025-03-06

The interview study continued with a structured grouping and analysis of results, with the aim of identifying, categorizing and visualizing the key findings from the qualitative data. With the proposed research questions in mind, the results were grouped into three categories:

- *The current practices related to weight management*
- *The causes of a weight delta*
- *The consequences of a weight delta*

The results were then analyzed and presented according to these categories, allowing for a clear distinction between how the system currently operates, where weaknesses lie, and the impact they have on the organization. Current practices within the organization were identified and presented in schematic figures. This approach helped with the overall understanding of the weight management processes and which part of the organization that are central actors or closely affected by its activities and results. The analysis of causes was conducted in the form of an Ishikawa, or Fishbone, Diagram (Ilie & Ciocoiu, 2010), which allowed a systematic organization of potential root causes of weight deltas across multiple dimensions, such as data quality, feedback-loops and human factors. Based on Kasser & Mackley's (2008) work, the consequences of weight deltas were structured in an impact model, highlighting main problems and pathways. The model helped illustrate the ripple effects of weight inaccuracies across the product lifecycle, highlighting how early miscalculations can cause inefficiencies and uncertainty later in the chain. The model helped to clarify which parts of the organization are affected and what types of working activities or decision making are disrupted when weight deltas occur.

3.3 Framework

The final portion of the project was the development of the framework. Based on the results found in the interview study, the outlined problem definitions and root causes were used to develop a framework. The framework was then validated through discussion with an internal expert, as well as through simulations using existing company product structures.

3.3.1 Development

Requirements on the framework were outlined based on interview findings and feedback from the industry supervisor. This was followed by brainstorming of potential solution approaches to meet the outlined requirements, conducted in an open discussion format in collaboration with the industry supervisor. The selection of solution approach was performed through a Weighted Decision Matrix as it was seen as a simple and straight-forward way of outlining and voting on relevant criteria (Odu, 2019). The criteria were stated based on interview findings regarding process considerations as well as the general project scope. The criteria were then weighed on a scale of 1-5, where 1 was deemed unimportant and 5 as very important. The voting was also completed on a scale of 1-5, where 1 was the lowest grade in terms of importance and 5 the highest. The voting was completed individually by the thesis workers and industry supervisors, with the final set score for each criterion being the mean of all votes. The weighted score for each criterion and potential solution approach was found by multiplying the voted score for a criterion with the criteria weight to achieve a weighted voted score. The total score for each potential solution approach was found by summarizing all weighted scores for each solution approach. The solution approach with the highest total score was selected to proceed with. The development of the solution approach was then completed in collaboration with the industry supervisor, mostly carried out through discussions held in online meetings. Through frequent feedback sessions, the framework was iteratively improved and expanded until the outlined requirements were deemed to be met.

3.3.2 Validation

The validation of the framework was conducted through an expert discussion with the industry supervisor, an internal stakeholder related to the weight management process at Volvo Cars. The framework proposal was examined and discussed where the framework's potential to meet outlined requirements was used as a guideline for validation. Further, the framework was manually applied to a sample of existing structures within the company software. This was done by selecting a function group from a current vehicle model and manually extracting the needed information. By mimicking the logic of the algorithm to existing structures in the generic BOM, it was ensured that a correct output was found.

3.4 Method Criticism

In general, the method used supported the purpose of the project well and allowed for successful study and framework development. Initially, the idea of benchmarking was proposed to better understand industry-wide perspectives. However, this was disregarded as the timeline of the project was quite short and no interest was shown from contacted firms.

4 INTERVIEW FINDINGS

In this section the results of the qualitative interviews are presented. The results are structured as an outline of three main areas: current processes and practices, identified causes of a weight delta and lastly, consequences of occurred weight deltas.

4.1 Volvo Cars' Current Practices

The current processes at the case company related to the weight management system lay within the two-layer configuration system. This system manages both parts and variants to meet internal and external needs and requirements.

4.1.1 Two-layer Configuration System

Volvo Cars work with a rule based, open configuration split into a feature domain and an item domain, such as the one presented by Tidstam (2014). The configuration system specific for Volvo Cars uses two layers, named the variant layer and part layer, shown in Figure 4. When a vehicle is configured, a *vehicle variant* is defined as complete configuration. Relating to the product structure presented by Voronov (2013) and Tidstam (2014), within Volvo Cars, feature families are defined as *variant families* and feature variants are described as *variant designations*. The complete unique vehicle configuration typically consists of one variant designation from each variant family and is defined by around 500 variant designations. The purpose of the variant layer is to more efficiently define a complete vehicle on a more abstract level than physical parts, both to meet customer needs on features and functionality, as well as internal requirements regarding configuration time.

The part layer consists of part data, stemming from designers and suppliers. This part data is in the form of characteristics such as materials and weight, as well as geometric data of part geometry in the firm's CAD software. The part data is stored in the PDM system, with the weight and geometrical data being used in the weight system. The part layer and the variant layer are connected through *variant expressions*, a form of Item Usage Rule that outlines for which variant designation or combination of variant designations a part applies.

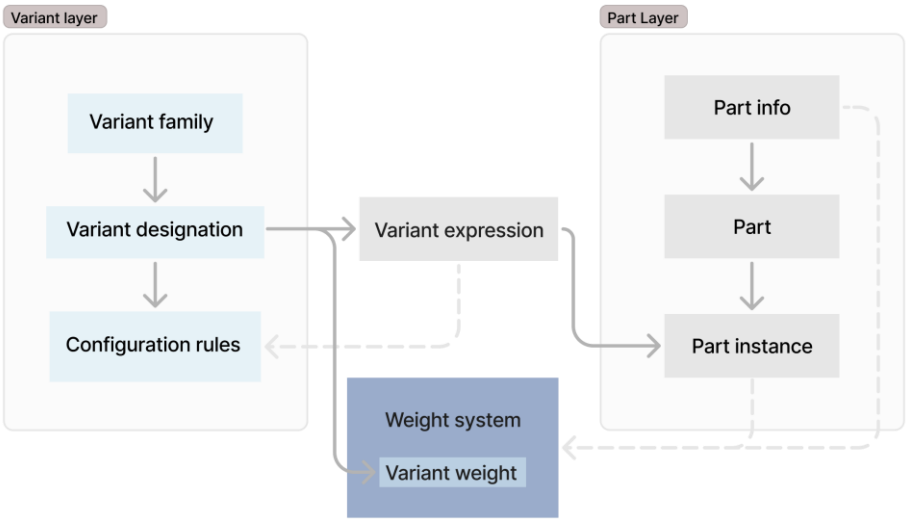


Figure 4. Two-layer configuration system.

Within the part layer the generic BOM is stored, which is used to organize all the parts and assemblies that are required to manufacture each specific product configuration. To manage the complexity of product variants and part information, Volvo Cars uses a hierarchal position (POS) structure to organize the generic BOM. The illustration in Figure 5 below outlines the structure of the system, and in what way it supports configurability and traceability. At the top level the generic BOM is divided into *function groups*, they represent major functional areas and systems within the vehicle, segmented in logical architectural categories. Each function group is then divided into POS codes, where each POS represents a specific physical or logical position in the vehicle. Each POS contains several POSTs (Part instances) which represent one instance of a unique part that applies for a certain configuration of that POS.

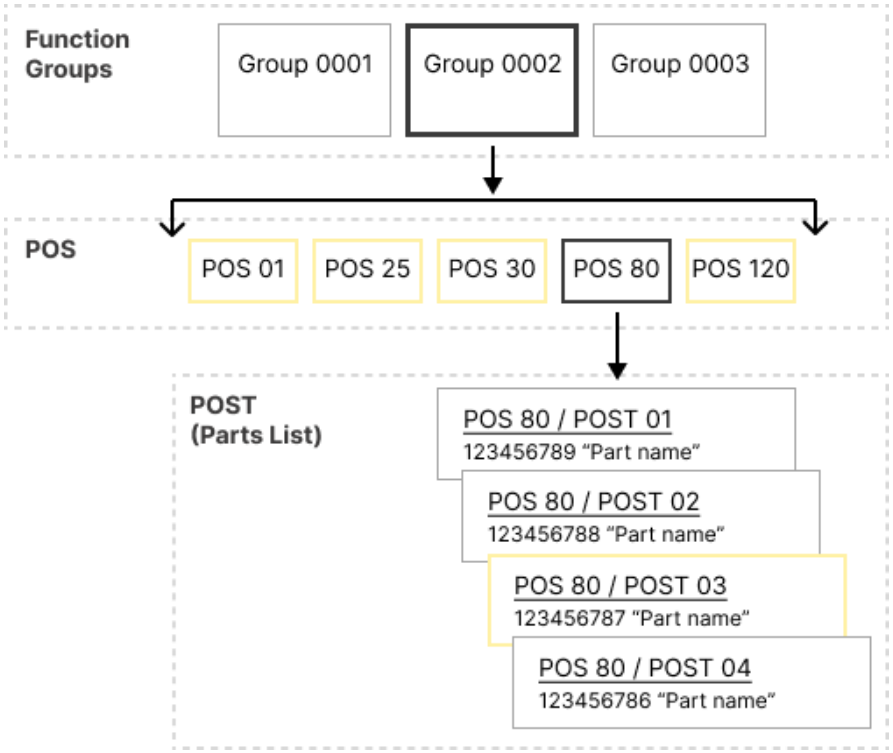


Figure 5. An illustration of the POS classification within a function group.

In this illustration, the function group is *Group 0002*. Within function Group 002, there is *POS 80*, where several part variations are listed as POST entries, each of them corresponding to a specific type of the part. In this case the figure illustrates that there are 4 different types of the part which are available to choose from (Volvo Cars, 2024).

4.1.2 The Variant Layer

When managing vehicle variants and their unique configurations, a systematic structure is used, illustrated in Figure 6. The figure outlines the relationships between variant families, variant designations and vehicle variants.

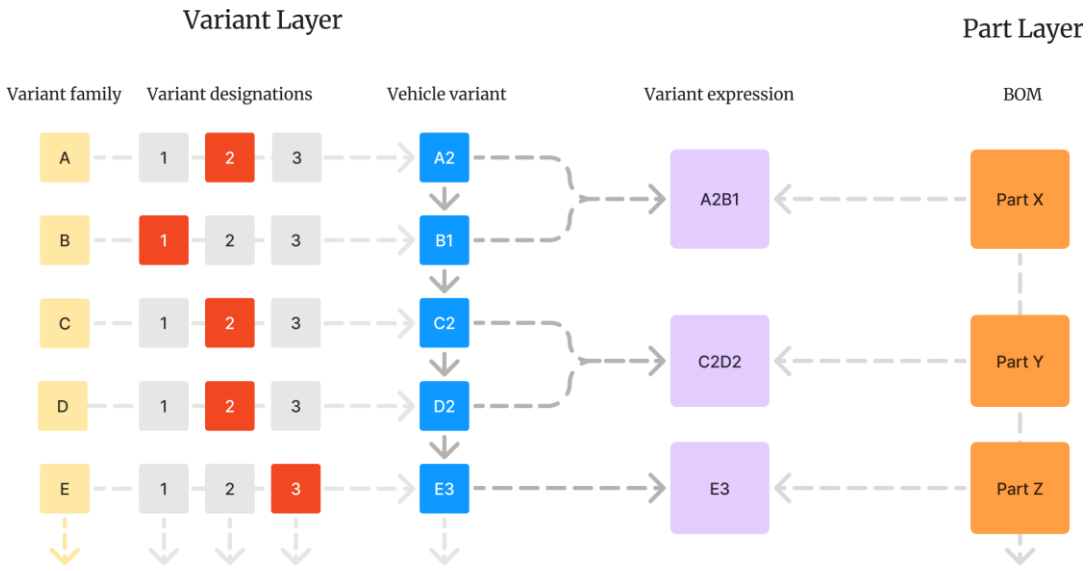


Figure 6. An illustration of how products are structured to support configuration.

The connections between the variant layer and part layer are the variant expressions, which describe the set of variant designations required for a specific part to apply. The variant expressions use the *if-then* structure described by Hayes-Roth (1985) to filter out which parts to use in a given configuration. Within the variant layer, there is a distinction beteween variants which are customer facing, often high-level features of the vehicle, and technical variants, which are lower level features used in production and product development. The *if-then* structure is also used here, as rule based configuration is utilized to filter out technical variants based on the selected customer variants. Within Volvo Cars, customer variants can be seen base variants, selected initially in the sales configurator, or as options or packages added to or subtracted from the base selections. Typical base variants include for instance driveline, regional market and exterior colour, together forming a base vehicle.

The creation of variant designations stems from a collaborative effort between designer engineers and product specification, with two different pathways as a result. Either new variant designations are created to assign parts created by designers, or parts are designed to technically solve a feature requested by the product planning, market or the actual project itself. When variant designations and configuration rules are created, Change Order specialists create variant expressions to define for which criteria certain parts apply. When parts are designed in response to a created variant, the part needs to be defined for the variant expression to be completed. The parts which are related to a variant may change during the lifecycle of any vehicle. This is done through a change order, or CO, dictating which parts are to be substituted, changed or added and at what time. The need for CO can come from several stakeholders in the process, often being related to supplier or market needs. As parts are added or modified, the weight and weight distribution of a given vehicle variant can therefore change over the course of its lifecycle.

4.1.3 Current Weight Management Process

The current weight management process at Volvo Cars relates to the work of documenting and assigning weight values to selected characteristics of a vehicle. Based on the product structure, variant expressions and part data, weight is manually assigned to arbitrarily chosen partial configurations. This part of the weight management process is referred to as *weight abstraction*, which enables the sales configurator to efficiently calculate the total weight of a specific vehicle configuration. This is done by summarizing the weight of the variant designations included in the individual vehicle's configuration. The total weight

is at the same time distributed between the front and rear axles, based on the calculation of the vehicle´s center of gravity. The calculation is entirely dependent on the customer´ s configuration choices, meaning that each unique combination of selected features influences the predicted weight outcome. Because each vehicle is unique, and the number of possible individual complete configurations is extensive, it is not feasible to predefine and store every possible vehicle variant. Instead, configurations are dynamically generated as an integral part of the configuration process, which also impacts how weight is determined:

"Due to the vast number of possible configurations, each sellable variant is not stored in the PDM system, it is calculated in the order-to-delivery process. Hence, the weight of a vehicle variant needs to be calculated as well" - (*Subject 1*)

The weight management process is made up of four distinct sub-processes, each playing a role in ensuring the overall quality and reliability of the weight predictions. Figure 7 presents a schematic overview of the weight management process as it looks today. As shown, the first step of the process is to define the part weight, which is gathered from suppliers or internally and then specified in the PDM system as a part of the product documentation process. Secondly, a generic BOM is created based on parts and their variant expressions. Data and information of design structure is then used to distribute weight over the front and rear axles of the vehicle.

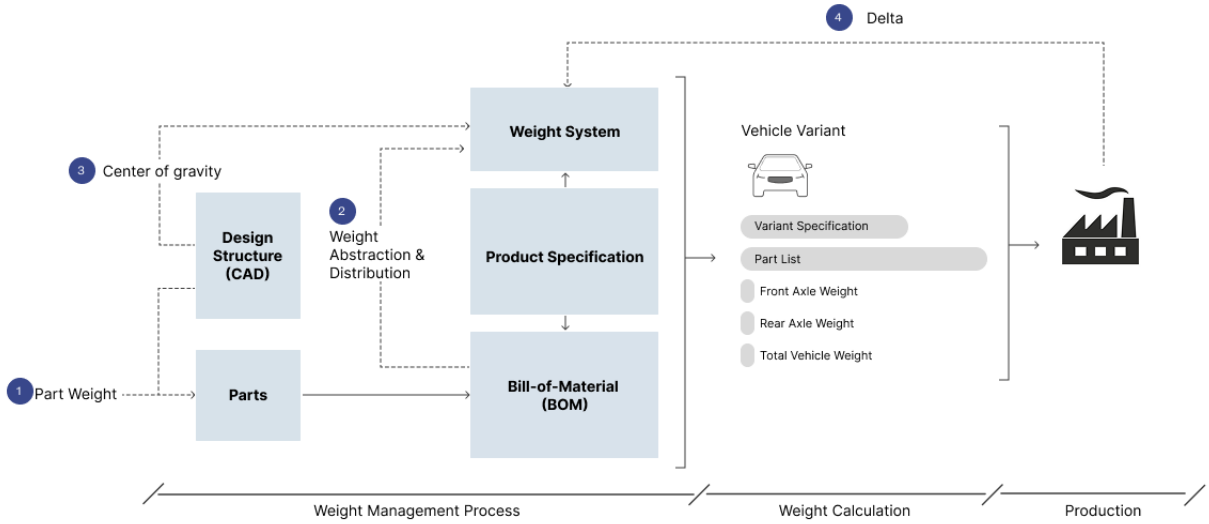


Figure 7. An overview of the weight management process.

The weight distribution for a certain vehicle configuration cannot be calculated automatically due to complexity in the connection between parts and variant designations, meaning that it today is a manual assessment. Weight is therefore manually abstracted and distributed from parts in the generic BOM to variant designations in the weight system. The contribution to total weight and axle distribution for arbitrarily chosen partial configurations is defined and managed in the weight system. As instances of vehicle variants are produced, the occurrence of a weight delta between predicted weight and actual weight is used to calibrate the weight system, through a feedback-loop. Usually, the weight system is prepared around 20 weeks before the first Virtual Prototype (VP), series begins. The VP is the first time physical vehicles are produced. These vehicles and other pre-series builds are used to calibrate the system, where feedback from weight delta measurements is sent back to weight engineers to improve

accuracy. However, the pre-series often differ from final customer configured vehicles in terms of parts and variants, and late changes to design and packaging are common. This makes the system challenging to maintain. Furthermore, not all vehicle variants are built during the pre-series phase, limiting the ability to fully verify the weight prediction across all configurations.

Lastly, it was explained that modifications in the final stages of vehicle development occur rather frequently, due to supplier constraints, performance adjustments or new regulations. Such changes cause a ripple effect that impact areas such as the generic BOM, configuration rules and vehicle's weight distribution. These impacts often require manual adjustments, causing inefficient and non-value creating work activities. Invalid configurations and delays in such processes also have a negative impact on production planning, as they slow down order processing and material allocation.

4.2 Causes of a Weight Delta

The occurrence of discrepancies between a vehicle's predicted weight, measured in kg and calculated in the weight management system, and the actual weight measured at the end of the production line, is referred to as a weight delta. The identified causes of a weight delta are shown in the Ishikawa Diagram in Figure 8 and described in detail in the following section.

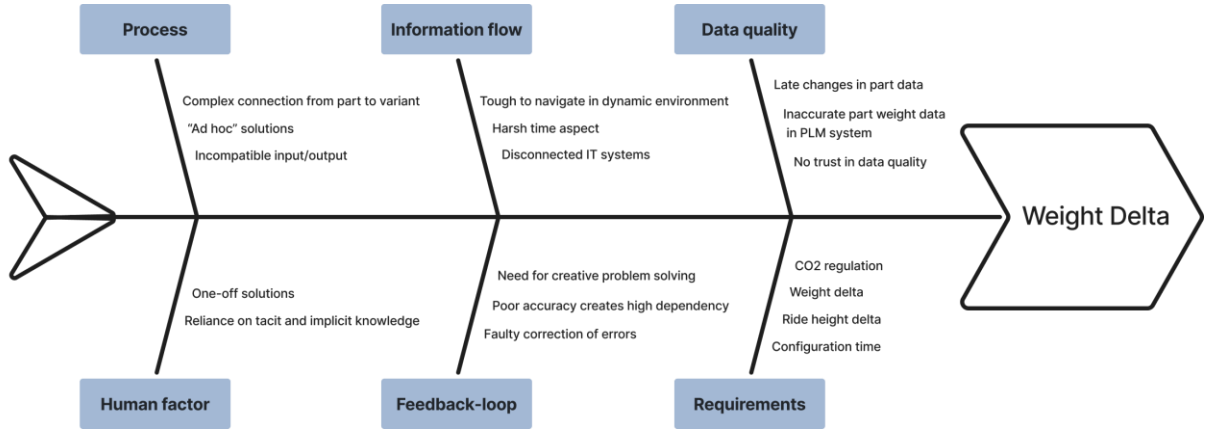


Figure 8. Identified causes of a weight delta.

4.2.1 Process

In the current structure of vehicle configurations at Volvo Cars, the generic BOM describes the physical structure of the vehicle, while variant designations typically represent functional or more customer-facing features. This means that the link between vehicle parts and variants is not always direct and easy to grasp. The occurring difference in structure causes that a single part can apply to multiple configuration scenarios, and a single variant designation can include several parts. The setup results in so called *many-to-many relationships* between physical vehicle parts and abstract variant designations. These complex relationships occur already in the first step of the weight management system. Since there are more parts than variant designations, it is not sufficient to simply transfer the weight of a part to the variant designation it appears to be linked to. Instead, it is necessary to understand the configuration context in which the part is used, only then can the weight be abstracted to the appropriate level in the system.

The data which is handed over from one function of the organization to another is often incompatible with the format best suited for the recipient. For instance, CO Specialists hand over variant expressions, consisting of partial configurations linked to parts, to the weight team. This is done as several variant designations often are required to define the condition of the variant expression. The weight team is in turn required to solve many-to-many relationships, complex systems of equations which sometimes do not have a correct single solution. An illustrative example of this type of challenge was provided from the interviews, see Figure 9. The illustration shows how different combinations of variant designations (represented by 1, 2, 3 and 4 in the figure) are described by the same parts or sub-assemblies (represented by A, B, C and D in the figure). It was explained that these types of relations require manual assessment to find the most suitable solution to where the weight should be assigned. The structure was described as an inconsistent system of equations, where there is no existing single solution. In this example, 10 kg can be assigned to the combination 1 and 3, but since variant 3 can affect a whole other set of parts as well, this weight abstraction is not obvious. This results in manual abstraction of weight data that is based on heuristics, expertise and engineering judgement, seen as ad-hoc. Even though a majority of the parts have simple variant expressions, there is still a significant portion of time spent on finding suitable solutions to these many-to-many relationships. As the solution then is a manual assessment of what is best suitable, the process is time consuming with a high risk of inaccurate weight prediction, which leads to deltas in predicted weight versus actual weight.

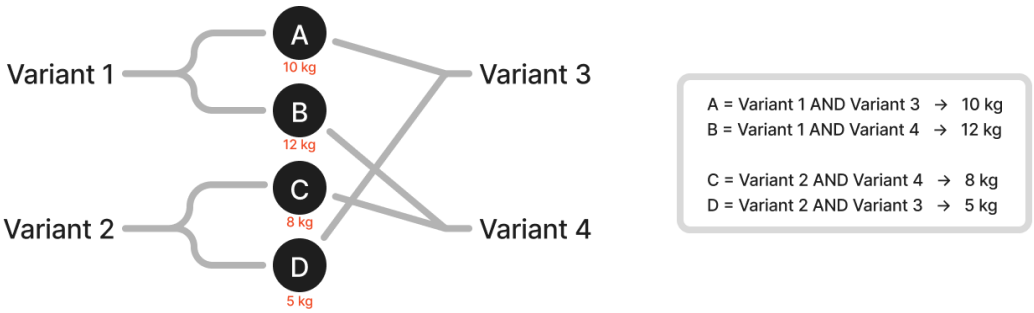


Figure 9. An illustrative example of many-to-many relationships in configuration scenarios.

4.2.2 Information Flow

The identified challenges regarding the flow of information within the organization begin with the dynamic environment. Due to the size and structure of the organization, together with continuous product development and production, the alignment of departments and internal stakeholders becomes challenging. This is further pronounced due to late product changes and harsh time requirements on steps in the development process, for instance regarding information deadlines, which complicates the flow of information cross-functionally. Further, the connection between systems which carry weight data is lacking, leading to the use of manual judgement and input. Subject 1 highlighted the issue of a lack of connection between the CAD and PDM systems and the resulting need to manually assign center of gravity and weight distribution on front and rear axles. Subject 6 substantiated this with the following quote:

" Today there is a missing connection between the system which can show the weight distribution of the part and the PDM system where the part weight is stored. There is no connection between the two, and that is one of the biggest problems we have. […] The result is a need for manual judgement, where is this cluster of parts in the car? The manual judgement is a manual judgement and can of course be a source of errors." - (Subject 6)

4.2.3 Data Quality

The responsibility of part data, including its weight, lays on the designer and is based on CAD software. This information is verified with part suppliers and then updated by the designer in the PDM system. Due to this extra verification of part data from suppliers, issues arise regarding data quality. The verification of part data can come in at a late stage of product development, creating issues for employees throughout the organization who are dependent on the existence and accuracy of said data. The late changes are not communicated well throughout the process, increasing the risk for discrepancies between used and accurate data. Further, the PDM system requires an input of certain part data, for instance weight, forcing designers to add inaccurate or incomplete data to parts to submit them in the system. This inaccuracy is not displayed in the PDM system, causing confusion and use of inaccurate data later in the weight management process, leading to faulty predictions.

4.2.4 Human Factor

Several steps of the weight management process, including the weight assignment to parts and variants as well as the definition of variants and variant expressions, are manual tasks conducted by employees. This manual aspect means that the process, and thus the resulting prediction, is dependent on the judgement and expertise of involved engineers. The dependency on this tacit and implicit knowledge within the firm has led to a vulnerability to personnel changes, seen in the weight team. There, personnel which were experienced and familiar with processes and ways of working left their positions, meaning that new personnel need longer time to do work of similar quality. Also, the lack of standardization and automation within the process leads to work that is completed case by case, in a one-off solution manner.

4.2.5 Feedback-loop

The current ways of working with weight management within the organization requires the need for consistent correction of errors. The low reliability within the processes creates the need for working with a type of feedback-loop, where each correction often is unique and requires manual work and creative problem solving. Deltas that have occurred in weight management process are often identified and corrected through different types of such feedback-loops, where the corrections are often carried out in a type of trial-and-error process as there is a lack of structured data and clear parameter relationships. It is the difference between predicted weight and actual weight that serves as the indicator for these corrections, where engineering expertise and tacit knowledge often is needed to identify the cause of weight deltas.

The feedback-loop in the current weight management process exists as a reaction to the occurrence of a weight delta. Rather than being part of a proactive system, it functions as a compensatory mechanism to correct errors after they have already occurred. Throughout the interviews, it was argued that the weaker and more inconsistent the process is, the more dependent the organization becomes on feedback-based corrections, and the more resources and expert knowledge are required to manage them. Furthermore, incorrect corrections can lead to even larger weight deltas, creating a compounding effect over time. It was discussed that reducing reliance on this type of process would require improvement of the traceability of configuration parameters and understanding how to perform corrections, and whether doing so will shrink or enlarge the weight delta.

4.2.6 Requirements on Current Processes

The weight management process is measured against several internal and external requirements. The internal requirements on the process are related to sales configurator performance, ride height of the vehicle and weight distribution on the front and rear axles. The ride height of the vehicle, see Figure 10, is related to the choice of springs used, which is dependent on the load on front and rear axle. The springs are divided into weight classes which will correspond to the correct ride height of the vehicle.

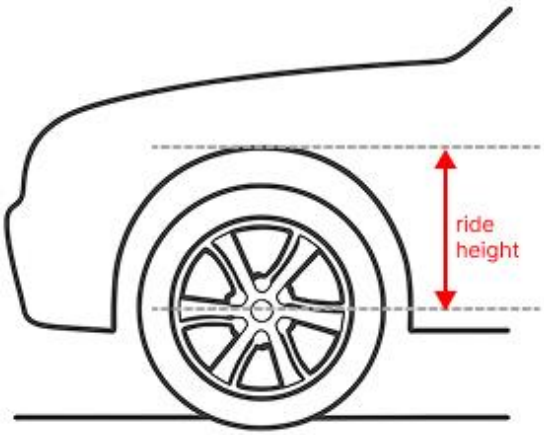


Figure 10. The ride height of the vehicle measured between the center of the wheel hub and fender arch.

The time requirement for the sales configurator was set to ensure smooth configurator experience for prospective customers wanting to choose features on Volvo Cars' website. Due to the complexity of a vehicle, thousands of individual parts are included in each vehicle variant. A result of this complexity, in combination with the tough time requirement, has been to move away from configurating vehicles in the generic BOM, as the vast amount parts prohibit a timely configuration. The BOM for the vehicle variant is not created until later in the process, before the vehicle goes into production. The time requirement for the sales configurator, as well as the need of knowing which springs and dampeners to use to meet weight and ride height requirements, has thus led to the need of predicting the vehicle weight in the variant layer in the configuration process. The internal requirements are proprietary, but are in a range as follows:

- *Configuration time ≤ in the hundreds of milliseconds*
- *Ride height Δ ≤ single digits mm*
- *Weight Δ ≤ single digits kg per axle (rear and front)*

The external requirements of the process are related to vehicle weight and performance. There is a maximum allowed weight based on government regulations, as well as performance regulations on fuel efficiency and range which have tax implications (Skatteverket, 2024; Transportstyrelsen, n.d.; Carlson et.al., 2013).

4.3 Consequences of a Weight Delta

Within the current processes, the occurrence of discrepancies between predicted and actual weight has consequences throughout the organization and value chain of the product, shown in Figure 11. Three avenues of consequences were identified as results of an occurrence of a weight delta through the interviews.

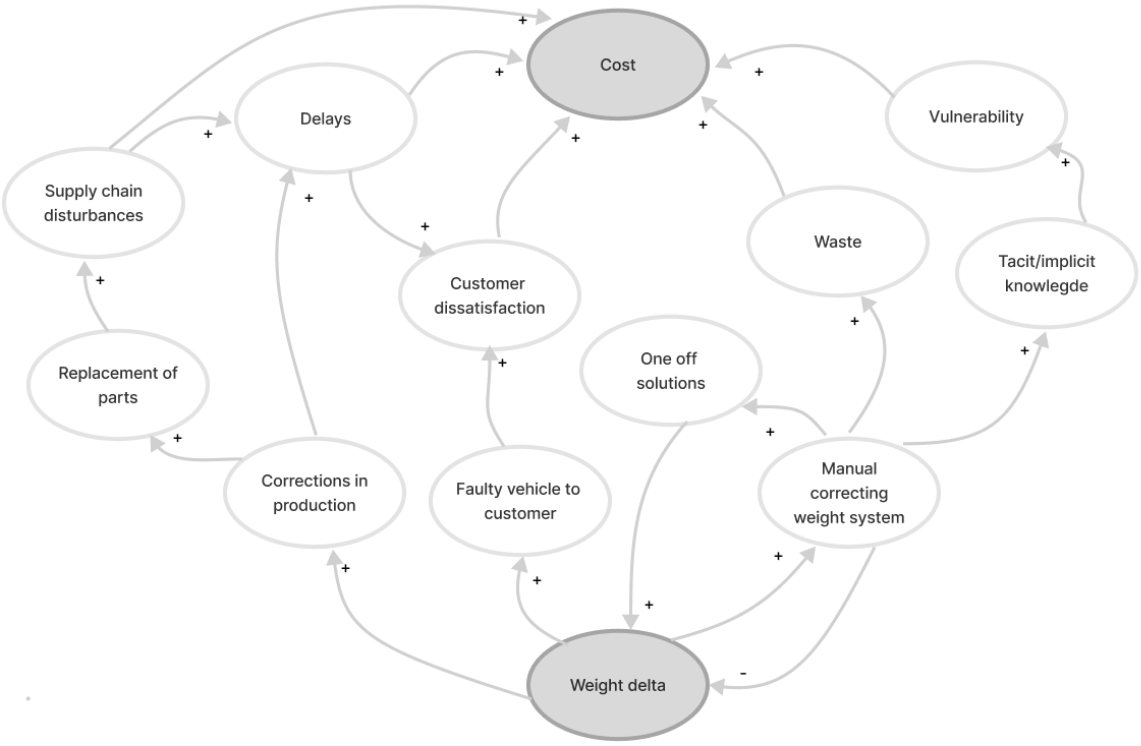


Figure 11. Impact model for discrepancies in predicted versus actual weight of vehicles.

Firstly, a correction of the vehicle is completed when a weight delta is found during, or with short notice to, the production of the vehicle. The correction consists mainly of replacing dampening and suspending parts to match the actual weight of the vehicle. This is a time-consuming activity, resulting in high costs and delays of production. Further, in the current production practices, any part which is used in production is intended for an ordered vehicle. When parts are replaced with short notice, these need to be re-ordered from suppliers with significant lead times, causing changes in production sequence or delays for vehicles intended for production at a later stage. This results in further delays, incurring greater costs.

Secondly, if a weight delta occurs and is not identified, a vehicle not matching the regulatory or company standards in place is delivered to a customer. This potentially leads to customer dissatisfaction and damage to the brand and reputation. Such an event would negatively impact future revenues, leading to costs for the firm.

Lastly, the occurrence of a weight delta currently results in a large amount of manual labor for employees. This manual labor consists of extra communication and problem solving. As this manual labor doesn't improve the quality of the product, it can be viewed as non-value adding, resulting in a cost for the firm. Further, this manual labor is often in the form of one-off solutions, not addressing root causes of the weight delta. As solutions often are batch specific, an adaptation which is applicable to

one batch may not address weight deltas in another batch and may cause greater or more discrepancies in future batches. The use of manual labor has also created a vulnerability to personnel changes within the organization, where current employees become very skilled in handling weight deltas within current communications and software systems. This tacit knowledge is hard to define and teach to new hires, meaning that the firm becomes quite dependent on retaining current personnel. When personnel retire or move to other organizations, the firm incurs costs as new hires are less efficient in the manual labor related to weight delta.

There is some manual labor which is aimed at decreasing future weight deltas, such as finding patterns of discrepancies for specific parts or configurations. Whilst this work can be successful in decreasing future weight deltas, it increases the dependency on current personnel within the organization, leading to greater vulnerability.

5 ANALYSIS OF INTERVIEW FINDINGS

The key findings from the interviews are analyzed and discussed in this section, focusing on the current challenges and opportunities within Volvo Cars' weight management processes. Insights from representatives from various departments were gathered, focusing on how the system operates today, where main pain points occur, and what potential there is for improvement. These findings formed the basis for the framework approach later developed in this thesis.

5.1 Identified Challenges in the Current Processes

Several aspects have led to the need for a weight management system containing weight abstraction and weight prediction within Volvo Cars. Weight prediction is necessary to fulfill regulations regarding CO_2, as well as for vehicle design choices, such as springs and dampeners, which must be completed in the configuration process. The possibility to summarize all included part weights in that step has been disregarded due to the extensive complexity in the BOM, where over 10 000 parts are included in a vehicle variant. This has been assumed to lead to too great computational load (Cannas et al., 2020) given the tough time requirement on the configuration process. This has thus led to the use of the variant layer for weight prediction, resulting in the need for a weight abstraction process.

The current weight management system at Volvo Cars faces several key challenges that affect both accuracy and maintainability, especially in the context of managing highly configurable vehicles. One of the main difficulties identified was the *extensive manual effort required throughout the process*. Much of the weight assignment, abstraction and validation relies on engineering experience and judgement, making the system resource-intensive and sensitive to human error. As the configuration handles thousands of parts and millions of possible vehicle combinations, this issue becomes especially critical.

Another challenge identified in the interviews concerns the *complex relationships between parts and configurations*, mainly focusing on the part-weight aspect. It became clear that there is a many-to-many relationship between some parts and variant designations, which causes the process of abstracting weight to appropriate levels to be resource intensive. Determining the most suitable solution for weight aggregation in such relationships requires understanding and experience of the full configuration process, which results in a rather large amount of tacit knowledge within the organization. The process therefore has a steep learning curve for new engineers, it is moreover time consuming to establish good prediction, accuracy and working routines, and the system requires a high level of specialization and responsibility to operate effectively.

In addition, some issues were seen in the timing and reliability of available data. *Accurate weight data were sometimes not available* until late in the product development process, as many parts were still under design or subject to change. This delays the ability to generate reliable weight predictions in early stages, which is important for system calibration and homologation planning. Building on this, calibrating the systems when changes occur seemed rather complex, especially late in the development cycle. Frequent part updates, variant changes and new configuration rules require manual adjustments and re-validation of existing weight systems.

Further on, it was observed from the interview results that the *identified causes of weight deltas are linked to one another*. The time requirement set on the sales configurator has led to the need to implement a weight abstraction to the variant layer. The use of an abstract variant layer has in turn resulted in the current processes of complex problem solving and manual labor required to handle variant

designations and many-to-many relationships between variants and parts. The negative consequences of the manual labor in the process are shown in the use of ad-hoc and one-off solutions, as well as in the vulnerability of relying on tacit and implicit knowledge within the organization. Further, the resulting weight deltas within the process increase the reliance on feedback loops to calibrate future weight predictions. This highlighted the need for structured, systematic approaches to weight aggregation and the management of the variant layer to improve efficiency, but also for scalability and overall data quality.

5.2 Opportunities for Process Improvement

With the identified causes for deltas in mind, the possibility emerged of decreasing the occurrences of weight deltas by improving or adapting the actual process itself. As the difficulties in the process are a root cause of several of the other identified causes, the framework development needed to focus on addressing process adaptations and improvements. The internal and external requirements on the process were not up for change and were kept into consideration during the framework development as set requirements on the adapted process. Further, as the human factor had been identified as a source of errors, the possibility of standardizing or automizing processes was also to be investigated.

The manual labor which permeates the weight abstraction process was identified to be costly for Volvo Cars from several perspectives. The Ishikawa diagram showed how manual labor resulted in inaccurate weight predictions, while also leaving the company vulnerable to personnel changes. The Impact model further showed that the time-consuming activities, which are a consequence of inaccurate weight prediction, were seen as wasteful and costly in and of themselves. By investigating the standardization and automation of weight abstraction, shown as step 2 in Figure 7, there was a large potential for cost savings identified. On a technical level, a clear opportunity to apply some kind of algorithmic reasoning and automation to this process was seen, and such a solution may offer the highest potential in terms of impact for the given time frame of the project.

6 DEVELOPMENT OF FRAMEWORK

The development of the framework is presented in this section, consisting of the general approach, requirements and solution alternatives. Further, the selection of solution approach is discussed, where the chosen solution is presented alongside an in-depth and visual description.

6.1 Framework Approach

During the interview study, several problem areas related to weight management were identified throughout the organization. Many of these areas follow a general theme of lack of standardization and automation, where manual input results in inefficiencies and sources of errors, especially prominent in the weight abstraction process. The analysis of the results thus showed that the best approach to this framework was to provide a suggestion of how the weight abstraction process could be automated. Further, the aim of the framework was to address the complexity of managing variant-level configuration data for accurate weight prediction by introducing a logically structured solution. This included the development of an algorithm solution that enables efficient and traceable abstraction of part-level weight data to variant level weight predictions of complete configurations. Further, this includes describing input and output structures of the framework and its underlying logic, focused on applicability within the current data environment at Volvo Cars.

Attempting to address other smaller issues identified in the process was not deemed as relevant for this project. The identified problems that haven´t been addressed in this thesis work, along with their weaknesses and suggested improvements approaches for the case company, are described and discussed in section 8.2 Future Work.

6.1.1 Requirements and Assumptions

The proposed framework was designed based on insights from the interview findings and the best practices found in the frame of reference. When applied, the framework should enable the fulfillment of the following requirements:

- *Automatically abstract weight from parts to one or more partial configuration consisting of one or more variant designations*
- *Enable weight prediction to current range of configurable individual vehicle variants in the sales configurator*
- *Enable a weight prediction which fulfills existing requirements regarding weight delta tolerances, ride height delta tolerances, sales configurator time and CO_2 calculations*

Further, some assumptions were made and set as a base for the development of the framework. These assumptions were primarily based on observations of existing practices and system structures and insights gathered from the interviews. The final solution relies on these assumptions to ensure internal consistency and practical relevance. The key assumptions are listed below:

- Output of the framework is a positive or negative weight assigned to a partial configuration (P), consisting of one or more variant designations. The output is in the form of *if "condition" then "consequence"*, as defined by Hayes-Roth (1985), where the condition is a partial configuration, and the consequence is an assigned weight. For example: If (Variant A and Variant B and Variant C) then (add 20 kg). The partial configuration is a part of at least one complete configuration, while being smaller than a complete configuration. By defining the output in this manner, it is compatible with the current sales configurator algorithm, which thus does not need to be adapted.

- Input to the weight abstraction process is a set of variant expressions, or IURs, linking a partial configuration, consisting of one or more variant designations, to a part instance with a defined weight. For example: If (Variant A and Variant B) then (Part X [10 kg]).
- It is assumed that the configuration problem can be analyzed as a CSP, described in section 2.2.2, where one variant designation must be selected from each variant family, such that all configuration rules are respected, to achieve a complete configuration. Thus, it is possible to calculate and analyze the configuration space for a set of variant families and find all possible complete configurations which a partial configuration can be extended to.

6.1.2 Potential Solution Approaches

Based on the analysis of current processes and the identified challenges within, several solution approaches were initially explored. In the early stages, a range of ideas were discussed to explore different ways of improving the weight management process. With the system requirements and the scope of the project in mind, these ideas resulted in four types of potential solution approaches. The proposed alternatives all represent different strategies for addressing the key issues in the processes, and were used as a basis for discussion, comparison, and final framework selection. Table 2 below presents an overview of these four approaches, along with a brief description of each.

Table 2. Description of the four potential solution approaches.

Solution approach	**Short description**
Single variants	Aims to abstract the weight from the part layer to the variant layer by assigning weight to atomic partial configurations, meaning single variant designations, based on the inputs of variant expressions. By defining the weight of each variant designation, the sales configurator can summarize the weight of the complete configuration by summarizing the weight of each ingoing variant designation.
Generative grouping	Automatic grouping of non-atomic partial configurations, larger than single variants, but smaller than a complete configuration, to which it assigns a weight. The algorithm finds the groupings generatively, thus including artificial intelligence aspects. The sales configurator filters the applicable partial configurations for the complete configuration and summarizes the weight of those.
Base car	Proposes a weight abstraction from the part layer to non-atomic partial configurations in the form of a range of “base” cars for each model. The algorithm will define which variants are included as base variants, outlining each base car, as well as which are seen as options. The algorithm will further describe how the weights of the options are added or subtracted from the base car in the sales configurator.
Sub-systems	Abstracting the weight from the part layer to the variant layer by identifying partial configurations within sub-systems. The sub-systems leverage groupings within existing structures in the generic BOM. The output will be summarized by the sales configurator against the complete configuration.

6.1.3 Selection of Solution Approach

Each of the potential solution approaches presented above provides a structured basis for comparison and evaluation in terms of finding and describing the most suitable framework. To determine which approach was the most appropriate for the current process, a structured selection process was carried out. A list of criteria relevant for the proposed solution was compiled. The selection criteria along with their weight and description can be seen in Table 3.

Table 3. Description of the solution approach criteria and their set weight.

Criteria	**Weight**	**Short description and weight motivation**
Weight assignment possible for any sellable configuration	5	The solution's ability to assign weight to all complete and valid configurations which can be created by the sales configurator. It was seen as a critical characteristic of a potential solution.
Sales configuration performance	3	The solution's ability to create a weight system which supports the time constraint on the sales configurator. Later optimizations would be able to improve sales configurator performance, giving a medium importance of the criterion.
Compatibility with current input	2	The solution's compatibility with the current format of input to the weight management system. A high grade infers a low need of adaptations in preceding processes. Adaptations of inputs were seen as unwanted but not unrealistic, decreasing the importance.
Reliability and accuracy of output	5	The solution's ability to create a weight system which consistently assigns weight to complete and valid configurations within existing tolerances on weight delta. The quality of the output was seen as crucial for a solution to be implemented.
Time efficiency of weight system creation	4	The length of time it takes the solution to create a weight system for weight prediction. Due to an important time aspect in the development process, a high grade was given.
Maintainability	3	The solution's ability to be maintained throughout the lifecycle of the product. This is dependent on the necessary frequency of updates (dependent on chosen input type), the ability to individually update affected portions of the solution and the ability to manually verify updates. This was seen as important for implementation, but not crucial within the objective of this thesis.
Implementability in current IT systems	3	The ability of the solution to function within current IT structures and systems. This was seen as relatively important, but not crucial as some adaptations are within reason.
Level of automation	5	The solution's added automation to the existing weight management process. As this was seen as the main purpose of the solution, the criterion was given the highest weight.
Ease of development	4	The ease of development of the solution within the bounds of this project, as well as the future work needed for an implemented solution. This was seen as an important criterion.

The above criteria were then used to evaluate the four solution approaches to find the most suitable solution. The selection of solution approach was based on the results from the Weighted Decision Matrix, which can be seen in Table 4 below.

Table 4. Weighted Decision Matrix showing the different solutions graded against the nine criteria.

Criteria	Weight	Single Variants	Generative Groupings	Base Car	Sub-systems
Weight assignment possible for any sellable configuration	**5**	5	5	5	5
Sales configuration performance	**3**	2	4	3	4
Compatibility with current input	**2**	5	5	3	5
Reliability and accuracy of output	**5**	4	3	2	4
Time efficiency of weight system creation	**4**	2	3	3	4
Maintainability	**3**	4	3	4	3
Implementability in current IT systems	**3**	5	5	3	5
Level of automation	**5**	5	5	4	4
Ease of development	**4**	2	2	3	4
Total weighted score		129	131	115	143

The weighted decision matrix indicated that the most suitable solution would be an approach that focuses on weight abstraction through sub-systems in existing BOM structures. This approach scored highest overall, particularly great possibilities were seen in terms of time efficiency and implementation into current systems, Furthermore, it showed the highest potential in ease of development within the scope of this project.

6.2 Fundamental Terminology and Key Concepts

To ensure clarity and consistency throughout the framework description, core terms were defined, presented in Table 5. These terms represent key structural and logical elements within the product configuration processes, and they form the conceptual foundation upon which the framework was built. The definitions are based on both established literature in the field, along with internal terminology used at Volvo Cars.

Table 5. Core terms of framework.

Term	Symbol	Definition
Function group	-	A manually designated group of part instances within a vehicle model's generic BOM, representing a certain functional area of the vehicle
Variant family	-	A variable describing a feature or characteristic of a vehicle
Variant designation	-	A value which a variant family can take the form of. For a complete configuration, one and only one variant designation must be selected from a variant family
Variant expression	-	the steering of a partial configuration to a part instances in the generic BOM, also referred to as *item usage rule*, or *IUR*
Complete configuration	-	All selected variant designations from the included variant families, satisfying all configuration rules, in a complete configuration
Configuration rule	C	Rules which define valid combinations of variant designations in a configuration
Partial configuration	P	A configuration containing at least one variant designation which can be extended to at least one complete configuration, but does not define a complete vehicle
Weight carrying partial configuration	P^w	A partial configuration which is used in the summary of total vehicle weight in the sales configurator
Part carrying partial configuration	P^p	A partial configuration linked to an individual part in the generic BOM through an IUR

In addition to the core terms, the framework relies on a set of key conceptual objects; The input of the weight abstraction process is the complete set of variant expressions, or IURs, linking partial configurations to parts in the generic BOM through a format of *if* "partial configuration" *then* "part". These partial configurations are referred to as part carrying partial configurations, or P^p. The output of

the algorithm will be the complete set of partial configurations linked to a weight, referred to as weight carrying partial configurations, or P^w, through a format of *if "*partial configuration*" then "*weight*"*.

6.3 Description of Proposed Algorithm

The developed solution of this project resulted in an algorithm description. The approach was to enable automated abstraction of weight data, aiming to support a more efficient and accurate weight management process at Volvo Cars. The following section describes the outline of the algorithms purpose and structure, the underlying logical principles and the process describes with support of visual examples.

6.3.1 Objective

The objective of the algorithm was to automate the weight abstraction process, i.e. the assigning of part weights to the variant layer by grouping and assigning weights to P^ws. The sales configurator would then filter out P^ws matching the variant designations included in the configured vehicle variant and summarize the weight to find the total vehicle weight. The objective of the algorithm was to create a system for defining all P^ws, such that the complete set of P^ws was smaller than the complete set of P^ps.

6.3.2 Algorithm Logic

To demonstrate the idea behind the proposed solution, the algorithm's overall logic is described through a simplified code format. This serves to illustrate the stepwise reasoning for weight abstraction and mapping, based on the set structure for the input, along with the desired output. In Table 6, the general structure of the proposed algorithm is presented.

Table 6. Structure of the proposed algorithm.

For given function group

(Adding input)

- For each part instance:
  - Extract quantity, part weight, IUR and partial configurations P^p
  - Add P^p as *condition* and (part weight) × (quantity) as *consequence* to *input pool*
  - If the exact P^p is already in the input pool, add (part weight) × (quantity) to corresponding *consequence* (in case of doubles)

(Weight abstraction)

- From P^ps, create set of variant families with corresponding variant designations in function group
- From set of variant families, simulate all possible configurations of the function group. The simulation includes all variant designations in the input pool and satisfies configuration rules C, creating the *Function Group Configuration Tree.* Each "branch" in the tree is defined as a P^w.
- For each P^p in input pool,
  - Find intersecting set of P^w in the Function Group Configuration Tree
  - Add weight to each intersecting P^w
- When the weight of all P^ps in the input pool has been added to intersecting P^ws, add the variant designations included in the P^w as *condition* and weight as *consequence* to the *output pool*

Iterate for all function groups

6.3.3 Algorithm Walkthrough

The algorithm operates in two steps. For the first step, the information in the generic BOM is converted to match the necessary format of the algorithm. For a given function group, the parts, part weights and IURs are converted to input. This is done by adding the partial configuration P^p linked to the part through the IUR as a *condition* and the part weight as *consequence* to the input pool. As there may be duplicates of partial configurations linked to different IURs in the function group, the algorithm checks for that specific partial configuration P^p before adding it to the input pool. If a duplicate is found, the part weight is added to the consequence of the already included partial configuration. In the case of an OR condition in the IUR, for instance: *if* (Variant A OR Variant B) *then* (part X [10 kg]), the resulting addition to the input pool would be: *if* Variant A *then* add 10 kg, *if* Variant B *then* add 10 kg. An example of the input pool can be seen in Table 7.

Table 7. Example of input pool.

Input pool		
Part Carrying Partial Configuration (P^p)	Condition (if)	Consequence (then)
P^p_1	A2 AND B1	add 3 kg
P^p_2	C1 AND D1	add 10 kg
P^p_3	A1 AND B1	add 7 kg
P^p_4	C2 AND D1	add 14 kg
P^p_5	C3 AND D1	add 11 kg
P^p_6	A1 AND B2	add 2 kg
P^p_7	C3 AND D2	add 6 kg

The second step abstracts weights by transferring the weight of each P^p to one or several P^ws. First, the algorithm uses all variant families and respective variant designations within the function group, together with the configuration rules C, to simulate all valid configurations for that specific function group. This creates the Function Group Configuration Tree, a version of a feature described by Tidstam (2014), seen in Figure 12 below. The tree shows the configuration possibilities for the included variant families and variant designations. The different choices that can be made in the configuration result in weight carrying partial configurations, or P^ws, appearing as horizontal pathways in the configuration tree. For each function group, the resulting P^ws are defined as complete branches in the Function Group Configuration Tree.

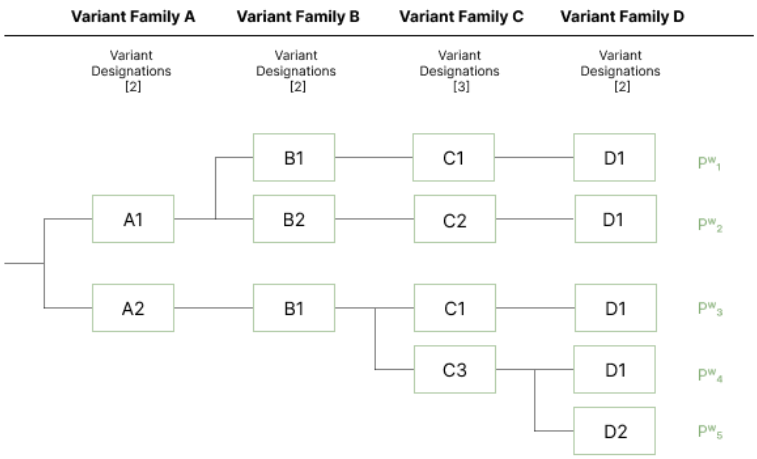


Figure 12. An example of a Function Group Configuration Tree, showing a simplified version.

The algorithm then iterates through the P^ps in the input pool and looks for intersections in the Function Group Configuration Tree. The corresponding weight of the P^p is then added to the intersecting P^w(s), as shown in Figure 13.

P^p_1: if **A2 AND B1** then **add 3 kg** to intersecting P^ws

Input Pool	
A2 AND B1	**add 3 kg**
C1 AND D1	add 10 kg
A1 AND B1	add 7 kg
C2 AND D1	add 14 kg
C3 AND D1	add 11 kg
A1 AND B2	add 2 kg
C3 AND D2	add 6 kg

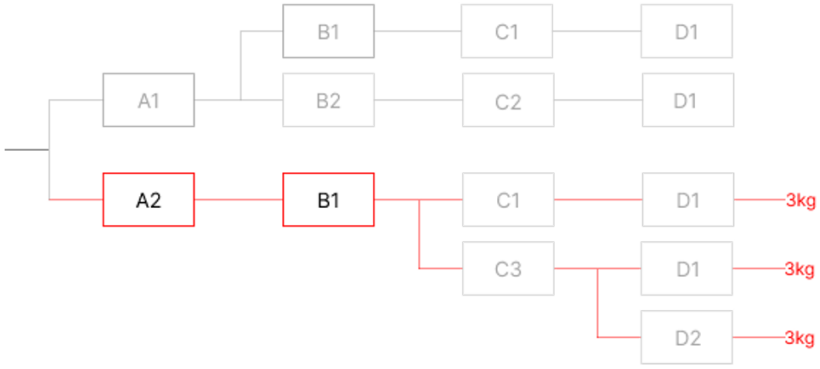


Figure 13. Illustration of how weight is added to the P^ws, branches highlighted in red, intersecting with P^p_1.

When the weight has been added to all intersecting branches, or P^ws, for that P^p, the algorithm moves on to the next P^p in the input pool. This can be seen in Figure 14.

P^p_2: if **C1 AND D1** then **add 10 kg** to intersecting P^ws

Input Pool	
A2 AND B1	add 3 kg
C1 AND D1	**add 10 kg**
A1 AND B1	add 7 kg
C2 AND D1	add 14 kg
C3 AND D1	add 11 kg
A1 AND B2	add 2 kg
C3 AND D2	add 6 kg

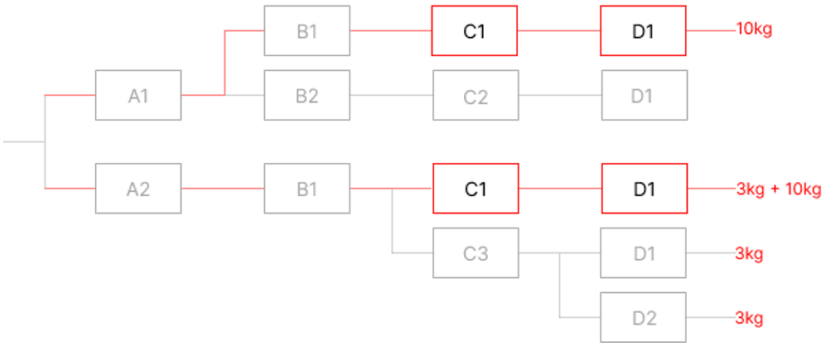


Figure 14. Illustration of how weight is added to the P^ws which intersect with P^p_2.

As the algorithm iterates through the input pool, weight is continuously added to each branch in the configuration tree according to each condition. To illustrate further, Figure 15 shows how the weight of 7 kg is added to P^W_1 according to the condition of P^p_3 in the input pool.

P^p_3: if **A1 AND B1** then **add 7 kg** to intersecting P^ws

Input Pool	
A2 AND B1	add 3 kg
C1 AND D1	add 10 kg
A1 AND B1	**add 7 kg**
C2 AND D1	add 14 kg
C3 AND D1	add 11 kg
A1 AND B2	add 2 kg
C3 AND D2	add 6 kg

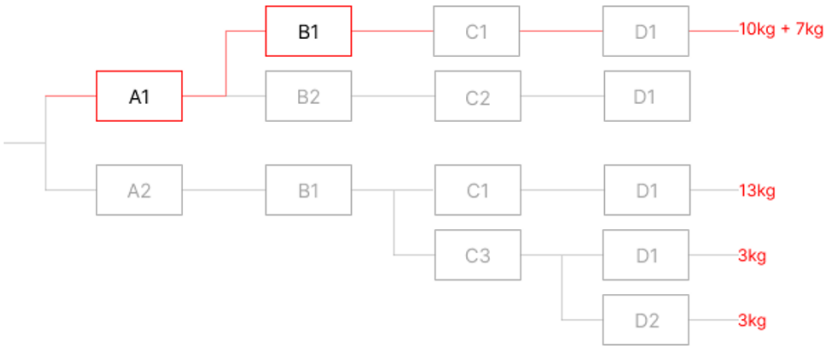


Figure 15. Illustration of how weight is added to the P^w which intersect with P^p_3.

When the weights of all P^ps in the input pool have been added to their intersecting P^ws, the function group is completed, and the algorithm moves on to the next function group. When all function groups have been iterated, the weight system is the complete set of P^ws and weights in the output pool. An example of all sets of P^ws for one function group can be seen in Table 8. In this example, the number of P^ws was 5 and the number of P^ps was 7.

Table 8. Example of output pool.

Output pool		
Weight Carrying Partial Configuration (P^w)	Condition (if)	Consequence (then)
P^w_1	A1 AND B1 AND C1 AND D1	add 17 kg
P^w_2	A1 AND B2 AND C2 AND D1	add 16 kg
P^w_3	A2 AND B1 AND C1 AND D1	add 13 kg
P^w_4	A2 AND B1 AND C3 AND D1	add 14 kg
P^w_5	A2 AND B1 AND C3 AND D2	add 9 kg

By evaluating the configuration problem as a CSP, a complete configuration must contain a variant designation from each variant family included in the function group (Voronov, 2013). Thus, as the Function Group Configuration Tree includes all variant families used in the function group, it can be guaranteed that one and one P^w only from each function group will be applicable for a given complete configuration and any complete configuration will match a P^w in each function group.

6.3.4 Assumptions and Limitations

A set assumption is that the computational time and storage capacity required to create the Function Group Configuration Tree is feasible and accessible. A limitation of the suggested logic is the exclusion of automated maintenance, meaning that continuous changes to variants, parts and constraints during the lifecycle of a vehicle model are not included in the weight abstraction system. A solution would have to be implemented to handle the dynamic nature of the product lifecycle to achieve maintainability of the weight management system. Regarding the assumption of the configuration being seen as a CSP, product structures which deviate from the definition of a CSP, meaning scenarios which deviate from one variant designation being chosen from each variant family, would need to be adapted for the solution to be able to be implemented. Thus, some verification of data structure may be required to fully enable automation at scale.

7 VALIDATION OF FRAMEWORK

To reflect on the proposed framework and evaluate its practical implications, this chapter discusses the validation activities that were completed, including both theoretical reasoning and simulation testing.

7.1 Validation

To support the credibility of the proposed framework, the development process was continuously iterated with validation efforts. Early on, discussions were held with industry stakeholders, particularly with the industry supervisor who is an expert within PLM systems. These discussions supported the definition of core assumptions and requirements, leading to guidance in the conceptual design approach. Further on, the dialogues helped the framework to be based on relevant constraints and internal system logic. Several of the conclusions drawn during this phase formed the basis for simplifying the problem formulation.

As a second stage of validation, the algorithm was simulated using a real product structure from the currently used PDM systems at Volvo Cars. The goal of this simulation was to evaluate how much automation and standardization in weight abstraction could be accomplished compared to the current method, and whether an increase in efficiency and accuracy could be achieved.

Together, the validation activities supported development of the solution and meant that the algorithm was compatible with existing input data and that the output was aligned with the requirements of the sales configurator.

7.1.1 Framework Simulation

To gain further insight into the potential of the framework, a simulation-based evaluation was conducted. A function group was selected from a vehicle already in production, where the raw data was manually extracted from the PDM systems. The logic of the proposed algorithm was then mimicked to ensure a valid and desirable output.

First, the raw data from PDM systems was extracted, as shown in Table 9. The format of A1, B1 functions in the same manner as A1 AND B1, which is the format described in 6.3.

Table 9. Raw PDM data.

Part	Quantity	Variant Expression	Part Weight*
Outer rear view mirror LH Euro/OS	1	LHD, LHMIR1	2.3 kg
Outer rear view mirror LH USA/Can	1	LHD, LHMIR2	2.6 kg
Outer rear view mirror LH India/UK	1	RHD, LHMIR1	2.4 kg
Flange M6 Screw	3	COMMON	0.2 kg
Outer rear view mirror RH Euro/OS	1	LHD, RHMIR1	2.3 kg
Outer rear view mirror RH USA/Can	1	LHD, RHMIR2	2.6 kg
Outer rear view mirror RH Arab	1	LHD, RHMIR4	2.5 kg
Outer rear view mirror RH India/UK	1	RHD, RHMIR1	2.4 kg
Flange M6 Screw	3	COMMON	0.2 kg

**Part weights are estimates*

The raw data was then converted into the format of the input pool, as shown in Table 10.

Table 10. Input pool for simulation.

Input pool	
Condition (if)	Consequence (then)
LHD, LHMIR1	add 2.3 kg
LHD, LHMIR2	add 2.6 kg
RHD, LHMIR1	add 2.4 kg
COMMON	add 1.2 kg
LHD, RHMIR1	add 2.3 kg
LHD, RHMIR2	add 2.6 kg
LHD, RHMIR4	add 2.5 kg
RHD, RHMIR1	add 2.4 kg

By applying relevant configuration rules, the Function Group Configuration Tree was created as shown in Figure 16.

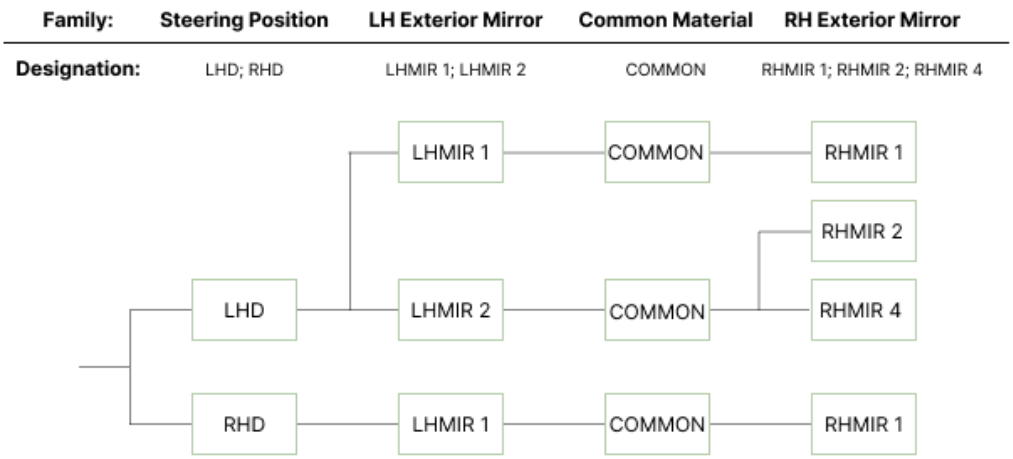


Figure 16. Function Group Configuration Tree for simulation.

By sequentially going through the input pool and finding intersecting $P^{w}(s)$ in the Function Group Configuration Tree, the resulting output pool was found as shown in Table 11.

Table 11. Output pool for simulation.

Output pool	
Condition (if)	Consequence (then)
LHD, LHMIR1, COMMON, RHMIR1	add 5.8 kg
LHD, LHMIR2, COMMON, RHMIR2	add 6.4 kg
LHD, LHMIR2, COMMON, RHMIR4	add 6.3 kg
RHD, LHMIR1, COMMON, RHMIR1	add 6.0 kg

7.1.2 Simulation Discussion

By simulating the proposed algorithm, 4 P^ws were found in the function group. For the studied vehicle model, there were 204 function groups. This means that the number of P^ws, or weight carrying elements, which the sales configurator will summarize for a given vehicle variant will be 204, as one P^w is selected from each function group during the configuration process. For the studied vehicle model, the median function group had 6 individual IURs with the largest function group having over 200 individual IURs. As shown in the simulation, which had 8 individual IURs, the median size of a function group is seen as manageable and could be expanded to a larger subsystem used for the proposed algorithm. However, the largest function group of over 200 IURs may be required to be split up into smaller subsystems to generate a manageable sized output pool, as each added variant family exponentially increases the size of the pool.

For the purpose of weight abstraction, the creation and storage of the output pool is the most important, as the algorithm will then store all P^w for the given function group. The configuration tree is a visual representation of the output pool but is not seen as necessary for the sake of successful storage of all P^w. Overall, the simulation showed great potential for automating weight abstraction and achieving efficiency improvements.

8 DISCUSSION

This chapter discusses the impacts of the proposed solution for the case company in relation to the found problem areas presented in Section 4, Interview Findings. Both opportunities for improvements and challenges for implementation are considered. Further, found problem areas which were not tackled by the developed framework are discussed as recommended future work. Lastly, the sustainability impacts of the proposed solution are presented.

8.1 Framework Impacts

The suggested framework for automating mass abstraction processes addresses key challenges identified at Volvo Cars. By introducing a structured method for abstracting weight from the part layer to weight carrying partial configurations, the framework provides a clear pathway for reducing manual workload and shortening the time for weight system creation. Further on, it should improve traceability since the structure of the algorithm has a well-defined link between parts and the Weight Carrying Partial Configurations (P^w). The solution also enhances scalability, especially as product complexity and the number of configurable features continue to grow. Further, since the foundation of the framework is based on existing product structures, such as function groups and variant families, implementation within Volvo Cars' current configuration environment is seen to be possible. Another important aspect which increases the potential to implement the proposed framework is that the input and output is compatible with the existing data flows within the organization. Overall, the design of the framework ensures compatibility while minimizing disruption to current processes.

A significant source of complexity identified in current weight management processes was the presence of many-to-many relationships between parts and variants designations. Such relationships make the process time intensive and dependent on manual interpretation. By systematically assigning weight according to the suggested algorithm, it ensures that each part weight is linked to logically defined P^ws. The issues which arise with attempting to manually deconstruct many-to-many relationships are therefore avoided. This automated approach would thus make the process less dependent on subjective decision making, as no manual labor is needed to create the weight system. It can be assumed that some manual effort is needed to manage the process and verify the result, however, this would be significantly less than what is currently need. Thus, notable cost reduction can potentially be achieved. Further, it can be assumed that both accuracy and operational efficiency when abstracting weight can be achieved with the proposed framework.

Moreover, further opportunities are seen in achieving higher efficiency with the proposed solution. By rethinking the underlying product structure used in the analysis, larger subsystems could be achieved. As variant families and designations overlap between function groups, there is a potential to merge function groups and create fewer but larger Function Group Configuration Trees. Fewer subsystems would mean fewer weight carrying elements for the sales configurator to summarize. By strategically defining these subsystems utilizing this overlap, the number of elements processed during configuration could potentially be lowered beyond what is achieved with the current proposal. This would help decrease the time required for weight prediction, however such an approach may introduce practical limitations. For instance, there will be an upper boundary to how large these subsystems can be before they become too computationally complex and difficult to manage. Having one single configuration tree for an entire vehicle model would result in excessive data size and performance issues, as all possible configurations then would be included. Therefore, there is a "sweet spot" to be found in the size of the

subsystems, where efficiency is high, but complexity is still manageable. The only demand on the sub-system is that each part instance belongs to one and only one sub-system.

Another opportunity for further improvement is the theoretical possibility to optimize the configuration step of the weight system. For a sequential configurator, the matching of criteria is done "top-down", meaning that the configurator filters partial configurations from the top of the output pool for each function group until it gets a match, and then moves on to the next function group. By placing more common configuration choices earlier, or higher up, in the output pool, there may be an increase in efficiency for the filtering of partial configurations. Further, in the case of duplicate P^w, these can be consolidated to create a more efficient output pool.

Despite the seen implementation possibilities of the framework, some aspects should still be considered to determine the full potential of the solution. To make the solution industry applicable, the dynamic nature of product structure, parts and the generic BOM throughout the lifecycle of the vehicle model needs to be considered, as the output pool will vary depending on different time intervals. This could be managed by treating time as an additional variable in the configuration tree, where "time" could be treated as a variant family and each time span as a specific variant designation. Further, as some parts in the generic BOM may deviate from the logic required for an applicable input to the algorithm, more extensive testing needs to be completed to ensure a comprehensive and valid input of all parts and variant expressions. Also, since the framework was built on a logical foundation, further verification would be required to assess its performance in handling data complexity and integration with existing IT systems. This is especially relevant regarding the weight distribution on front and rear axle, as the proposed algorithm doesn't take this into consideration and still requires manual input of weight distribution. Provided with a clearer connection between CAD and PDM systems, the output pool could take the form of: *if* P^w *then* add X kg to rear axle and Y kg to front axle.

A more extensive verification phase involving the development of a prototype or a programmed version of the algorithm is recommended to better determine the credibility and practical feasibility of the suggested solution. Moreover, a limitation is that the framework is built on the assumption that Volvo Cars formally continues to assign weight at part level. This foundational principle means that all weight abstraction and logic must originate from individual parts registered in the generic BOM.

8.2 Future Work

The development of the suggested framework for automated weight abstraction was based on a broad study of the weight management process at Volvo Cars. Due to the wide research areas of the study, a comprehensive view of the process and its challenges was developed. Despite that the framework provided a suggested solution approach for enhancing one main area of the weight management process, there were other areas that emerged during the research phase as well. These areas have not been fully explored in the framework development; however, value is seen in exploring these areas further for enhancing the overall reliability and efficiency.

The interview findings indicated that data quality within the generic BOM plays a foundational role in ensuring accurate weight predictions. Issues such as incomplete or outdated part weight data, lack of traceability and unclear ownership of data updates were highlighted. Recommendations for further investigation and improvements relate to establishing more control of generic BOM data by some type of status tracking and version control, along with more accountability of the data within the generic BOM. Enhanced traceability would help with linking weight entries to their source and current validity

as well. These improvements would significantly strengthen the integrity of weight predictions, reduce the uncertainty, and further on also enable faster troubleshooting.

Another area identified was the need to improve integration between the current IT systems being used throughout the organization. Fragmented data flows were seen in current processes, along with disconnected input and output between systems. These disconnections were assumed to hinder product documentation and weight prediction within the configuration processes. This area was also seen as one of the reasons for the need for feedback-loops between departments, as they are a result of trying to correct errors made in the weight systems. This identified fragmentation increases the risk of inconsistencies and efforts across departments. Synchronizing and integrating systems for product structure, such as PDM and CAD, would create a more reliable data environment and minimize the need for manual efforts.

Finally, this study has underscored the importance of cross-functional collaboration between different teams, including design, product specification and the weight teams. Many of the current challenges stem from siloed working practices, where changes in one area are not always fully communicated or understood by others. Establishing more formalized and collaborating workflows, along with more integrated IT systems, could foster a shared understanding of variant logic and process requirements.

While these recommended areas for future improvements extend beyond the scope of the framework developed in this thesis, they still are seen as essential for achieving sustainable improvements in the weight management processes. Parallel improvements in these areas, along with the development and implementation of the proposed framework, are expected to strengthen the company's capabilities to manage increasing product complexity and evolving requirements over time.

8.3 Sustainability Impacts

A successful automation of the weight management process promotes engineering resource efficiency due to the decreased manual labor needed in the process. This results in cost savings for the firm and less engineering resources being utilized to a large extent in value-creating activities. Further, improved accuracy of the system enables efficient use of material resources in production, decreasing material waste and energy usage. The improvements thus also promote ecological sustainability within the firm.

9 CONCLUSION

The main findings of the study are summarized in this chapter. Reflections on how identified opportunities and challenges can shape future improvements in weight management processes are also provided.

9.1 Summary of Key Findings

The activities of the thesis work gave many insights regarding the strengths and weaknesses of the current processes regarding weight prediction at Volvo Cars. The purpose of the thesis was to investigate the consequences of faulty weight predictions, identify factors inhibiting process efficiency, and explore ways to achieve more accuracy and scalability in weight management. Through a combination of interviews, document analysis, and theoretical development, key insights have been gathered that highlight both the complexity and the potential of weight prediction within the automotive industry. These findings are summarized below, directly addressing the research questions presented in 2.4.

What are the consequences of faulty weight prediction in an automotive manufacturer?

- Consequences of faulty weight predictions were shown to cause unnecessary costs for the firm, which accumulated in both direct and indirect ways. Direct consequences include incorrect part selection (for example springs or dampers), causing delays in production. Indirect consequences involve risks for performance deviations or regulatory compliance, such as faulty estimated emissions or taxes. These errors contribute to inefficient use of engineering resources and a decreased trust in data quality across departments.

What challenges inhibit an efficient weight prediction process in an automotive manufacturer?

- The inhibiting challenges were found to be mainly focused on a high reliance on time-consuming manual processes. Due to complex many-to-many relationships in the product structure and occurrences of inaccurate part data, creating the weight system was found to be a difficult and time-intensive activity. The system was also found to be sensitive to changes in product specification and product design, which complicate traceability and maintainability. A lack of standardized methods for abstracting part weights was seen as a cause of inconsistencies and dependency on tacit knowledge within the organization.

How can processes regarding weight prediction for unique vehicle configurations be refined to ensure greater efficiency and accuracy?

- Several refinement opportunities were seen in different areas of the weight management system, however the most accurate approach recommended for Volvo Cars is to introduce a systematic framework for weight abstraction. In such a framework, part weights are automatically added to configurations of sub-systems, defined as as “Weight Carrying Partial Configurations”, by taking advantage of already existing product structures. This approach would enhance efficiency and traceability, which in turn reduces the dependency on tacit knowledge within the organization.

9.2 Contributions

This project has identified challenges in weight management for automotive manufacturing, finding both causes to and consequences of faulty vehicle weight predictions. Further, a method for integrating weight prediction automation in the configuration of automotive vehicles was developed. Neither of these aspects were found in existing literature and can be seen as contributions to the field of Product Lifecycle Management (PLM) within automotive manufacturing.

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APPENDIX

Appendix A – Interview guide

Short introduction and context

We are students from KTH studying our last year in Innovation management and integrated product design. This semester we are writing our master thesis at Volvo Cars, focusing on improving the processes related to weight management.

General Questions

- To begin with, can you explain a bit about your work at Volvo Cars and which department you are in?
- How long have you worked at Volvo Cars/had this role?
- What is your role in the weight management process?
    - How often do you work with weight-related topics? What do those tasks look like?
    - Which systems or tools do you use to perform these tasks?

Current weight management process

- How does the weight management process work from your perspective?
    - What problems do you see with the current process?
    - What are the biggest challenges?
        - What are the most often occurring issues at your department? What are the consequences of incorrect weight management?
    - What limitations do you see?
    - Are there steps in the process that feel unnecessarily time-consuming or manual?
- Which departments or people do you collaborate with regarding weight management?
- How is weight data managed today? Do you consider it an efficient process?

Role specific questions

Weight management process

- Can you explain the general process regarding weight prediction as it is today?
- What are the main considerations for the weight management process?
- How well does the current system predict the final vehicle weight?
    - How common are discrepancies between estimated weight and actual weight?
    - Can you identify what caused miscalculation, and what parts of the vehicle that were affected?
    - What are the most common causes for miscalculation?
    - Are causes for discrepancies apparent throughout the organization?
    - Who is responsible for addressing causes of discrepancies?
- What factors contribute to uncertainty in weight prediction?
- How do you think that weight management processes could be more automated or standardized?
- Which steps do you think in weight abstraction and weight distribution could be improved through automation?
- How is distribution of weight on front/rear axles determined?
    - What are the tolerances for weight distribution on axles?
    - What challenges exist for distribution of weight on front/rear axles?
    - How common are discrepancies in weight distribution?

Product structure

- How are variants determined/established?
- How are the variant combinations determined?
    - Who is responsible for the determination?
    - How is weight assigned to variants and variant combinations?
- How are parts assigned to variants/variant expressions?
- When and how is weight assigned to parts?
    - How accurate is the weight data on parts?
- Can you describe many-to-many relationships in the product structure?
    - Which challenges are related to these relationships?

Problem areas

- What challenges come from assigning weight to variants and variant combinations?
- What challenges lead to quality issues in weight management?
    - What types of activities are most time consuming?
    - What are the requirements on the weight management process? (Time, accuracy etc.)
- What factors are dependent on correct weight prediction? (dampening/suspension, emissions/BEV range, regulatory)
    - What are the consequences of incorrect prediction?
    - worst-case effects/costs
- What are the impacts from change orders on the weight process?
    - Is weight automatically updated in a change order?
    - For parts and variants?
- How common is the occurrence of weight bearing variants/parts which do not impact weight calculations?

Possible solutions

- How could the assignment of parts to variants and variant combinations be improved to simplify the weight management process?
- Do you have specific ideas for improvements in the weight management process?

Wrap up

Do you have anything else to add? Or that you wish to discuss?

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